Understanding Criminal Record Penalties in the Labor Market

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Abstract

This paper studies the earnings and employment penalties associated with a criminal record. Using a large-scale dataset linking criminal justice and employer-employee wage records, we estimate two-way fixed effects models that decompose earnings into worker's portable earnings potential and firm pay premia, both of which are allowed to shift after a worker acquires a record. We find that firm pay premia explain a small share of earnings gaps between workers with and without a record. There is little evidence of variable within-firm premia gaps either. Instead, components of workers' earnings potential that persist across firms explain the bulk of gaps. Conditional on earnings potential, workers with a record are also substantially less likely to be employed. Difference-in-differences estimates comparing workers' first conviction to workers charged but not convicted or charged later support these findings. The results suggest that criminal record penalties operate primarily by changing whether workers are employed and their earnings potential at every firm rather than increasing sorting into lower-paying jobs, although the bulk of gaps can be attributed to differences that existed prior to acquiring a record.

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Workers with a criminal record have significantly lower rates of employment and earnings than workers without a record (Western, 2002). These gaps may be the obvious result of the barriers workers with a criminal record face in the labor market. Simply having a record, for example, legally precludes work in some occupations.[1] Even if not specifically prohibited from hiring them, employers may be reluctant to take on workers with a criminal history due to concerns such as perceived liability risk (Holzer, Raphael and Stoll, 2006; Cullen, Dobbie and Hoffman, 2023). Correspondence studies have consistently shown sizable callback penalties for disclosing a criminal record on job applications, even in entry-level positions paying near the minimum wage in retail and fast food (Pager, 2003; Pager, Bonikowski and Western, 2009; Uggen et al., 2014; Agan and Starr, 2018).

However, workers with a record frequently face a host of other challenges, including limited formal education and prior labor market experience. Roughly 65% of the incarcerated population did not finish 12th grade, for example (Harlow, 2003).[2] Prior evidence indicates that many individuals work infrequently and earn little even prior to their first encounter with the criminal justice system (Looney and Turner, 2018; Garin et al., 2025). Thus, it remains unclear how much pre-existing disadvantage shapes outcomes relative to the costs of navigating the labor market with a criminal record.

This paper addresses this question using a newly constructed dataset linking state administrative criminal justice records to earnings and employment data at the Census Bureau. Our criminal justice data has sufficient coverage to reliably measure adult criminal justice interactions for more than eight million workers. Unemployment insurance records allow us to track wage earnings and employer identifiers, while information from Census surveys adds additional detail on labor market activity such as occupation. The combined data provides an unprecedented opportunity to measure and unpack criminal record penalties in the U.S. labor market with a sample of 139 million person-quarter observations.

We begin by documenting the overall gap in earnings and employment between workers with and without a record. We focus exclusively on men, since women make up a small share of the population with serious criminal convictions, and pay special attention to felony convictions, as they typically carry the most severe potential labor market consequences. Workers with a

[1] For example, individuals with felony drug, weapons, property or other convictions are barred from serving as a TSA agent (49 U.S.C. §44936); convictions for "any criminal offense involving dishonesty or a breach" can be disqualifying for positions at FDIC-insured intuitions (12 U.S.C. §1829); drug and other convictions can also be disqualifying at any federally funded facility serving children under 18 (34 U.S.C. §20351). State-specific laws specify many other exclusions, including in licensed occupations (The National Conference of State Legislatures, 2024).

[2] Approximately 18% of the overall population 18 and older did not complete 12th grade at the time this statistic was computed.

felony conviction have about 11 percentage points (pp) lower employment rates than those without one. About 10 percent of these workers are in prison in any given quarter, however, potentially explaining some of these gaps. When individuals with a felony record have some earnings, they take home about 60% less than workers without a record. Workers with any prior conviction (not just a felony) show similar but smaller overall gaps, with 5 pp lower employment and about 40% lower earnings conditional on working. Difference in co-worker pay account for most of the earnings gap, suggesting that firms may play a critical role in explaining wage disparities between workers with and without a record.

To better understand the sources of these disparities, we estimate variations on two-way fixed effect models of earnings pioneered by Abowd, Kramarz and Margolis (1999) (hereinafter AKM). Our models allow log earnings to depend on a person-specific effect and a firm effect, both of which are interacted with criminal record status. This simple framework allows log earnings to decline after a conviction through sorting to new types of jobs (e.g., more low-paid retail work), declines in pay premia at the same firm (e.g., due to discrimination), and drops in person effects (e.g., due to productivity declines). By comparing estimated person effects with observed employment rates, we can then relate differences in the frequency of work to differences in earnings potential across worker groups.

We find that person and firm effects are important contributors to earnings dispersion among workers both with and without a record. After correcting for sampling error, the estimated standard deviation of firm effects amounts to a 22% change in earnings in both groups, for example. Person effects are roughly twice as variable as firm effects among workers without a record, but not among workers with one, consistent with the low average levels of skill in that population. There is no evidence of variation in within-firm pay premia gaps between workers with vs. without a record. The sampling error de-biased variance of gaps is in fact slightly negative. This finding suggests that if workers with a record tend to be paid less than similar workers without a record at the same firm, such a penalty does not differ strongly across firms.

Our estimates provide a simple way to decompose log earnings gaps between groups, building on the analyses in Card, Cardoso and Kline (2016) and Gerard et al. (2021), among others. Our approach leverages panel variation in record status to decompose the record penalty into differences in person effects that existed *prior* to acquiring a record, sorting across firms with different pay premia, and changes in earnings potential post record. This last factor captures either drops in person effects after acquiring a record or a within-firm record penalty that is common to all firms and workers with a record. While the relative role of these two factors is not identified, we investigate their potential contributions using assumptions commonly

considered in the literature.

The results show that the sorting component explains only 11% of earnings gaps. Differences in pre-record person effects explain a further 66% of gaps. The drop in earnings potential post conviction explains the final 23%. Assuming workers are paid their person effect in the restaurant industry, as is commonly done in the literature (Card, Cardoso and Kline, 2016; Gerard et al., 2021; Arellano-Bover and San, 2024), implies all of the drop in earnings potential reflects a reduction in person effects rather than an emergence of within-firm gaps in pay premia. Similar findings emerge when we re-weight the non-record sample to match the record sample's demographic, geography, industry, or occupation distribution, implying that sorting effects are a persistent but relatively small component of gaps even among similar workers in similar sectors and places. These findings imply that record penalties operate by changing earnings potential at every firm rather than through sorting, although differences that pre-date the record itself explain the bulk of the overall penalty.

Pre-record person effects are also related to differences in employment rates across groups. Using bias-corrected regressions of employment rates onto estimated person effects, we find that differences in pre-record person effects explain 30% of employment rate gaps between workers with vs. without a felony record and 42% of the smaller gaps between workers with and without any criminal record. Accounting for maximal drops in person effects post conviction, they can explain at most 37% and 49% of employment gaps for each record type, respectively. The size of the residual gap unexplained by person effects compared to sorting effects for earnings suggests that the challenges of navigating the labor market with a record primarily impact whether a worker finds a job at all instead of its overall pay level.

An important caveat to the results so far is that differences in sorting patterns between workers with vs. without a record may be attributable to some other, persistent feature of the worker (e.g., their skills or preferences at labor market entry). These features may have led workers to different types of jobs even if they had never acquired a record. In the final part of the paper, we analyze simple difference-in-differences specifications that leverage the panel dimension of our data to isolate and decompose the contribution of acquiring a criminal record itself. This analysis compares workers' outcomes around their first felony conviction to those of several matched controls, including similar workers who never acquire a record, workers charged but not convicted, and workers charged in the future.

The results show substantial shifts in outcomes around a felony conviction, with declines in employment of around 10 pp and log earnings of about 8 points. Difference-in-differences estimates are largest compared to matched controls with no recorded justice interaction, but

these workers appear to be on meaningfully different trends long before the conviction event. Compared to workers with future charges or workers charged but not convicted, difference-in-differences estimates point to smaller but still substantial earnings and employment declines. Increases in incarceration explains some, but not all, of the declines in employment.

The evidence that declines in firm effects contribute to the earnings impacts of a felony conviction is limited. Workers with no interaction with the justice system see significant improvements in firm effects over a comparable period, but relative to workers charged but not convicted or first charged in the future, the estimated impact of a felony conviction on average firm effects is less than one log point. This drop is both small overall and a small share of the estimated decline in log earnings post conviction. Instead, log earnings declines are explained by drops in earnings potential—either changes in person effects or the emergence of a common within-firm gap. Taken together, the panel evidence thus paints a similar picture to the overall decomposition exercises: *where* a worker is employed is a much less important determinant of criminal record penalties in the labor market than whether the worker is employed at all and their overall earnings potential at any firm.

This paper contributes to a large literature on the connections between the labor market and criminal justice system. The labor market disadvantage of individuals with prior criminal justice contact has been documented extensively (e.g., Western, 2002). Studies that attempt to isolate the role of criminal charges and convictions tend to point to large declines in employment and earnings around the event (Mueller-Smith and Schnepel, 2021; Rose, 2021; Agan et al., 2024*a*). Several analyses have also studied criminal justice interactions further upstream, such as arrest (Grogger, 1995) and pretrial detention (Dobbie, Goldin and Yang, 2018). A parallel literature shows that disclosing a criminal record substantially harms job application prospects (e.g., Pager, 2003; Agan and Starr, 2018) and studies employer motivations for disqualifying workers with criminal histories (Holzer, Raphael and Stoll, 2006; Cullen, Dobbie and Hoffman, 2023). Our analysis builds on this literature by studying the sources of criminal record penalties in the labor market and the contribution of employer sorting.

This work also connects more broadly to work on the overall impacts of criminal justice interactions, where evidence increasingly suggests that convictions (instead of sentencing punishments) prompt material changes (Humphries et al., 2024; Kamat, Norris and Pecenco, 2024). Holding conviction status fixed, for example, evidence suggests incarceration may reduce recidivism but have limited impacts on earnings and employment (Rose and Shem-Tov, 2021; Jordan, Karger and Neal, 2023; Garin et al., 2025). At the same time, other work suggests that undoing any harms of a criminal record may be challenging (Agan et al.,

2024*a*,*b*), consistent with persistent skills being the most important factor for outcomes. Our analysis supports the central role of persistent skills in driving labor market outcomes for workers with a record.

We also contribute to a growing literature using two-way worker firm fixed effect models to study earnings inequality within and between groups. These analyses have studied gender gaps in Portugal (Card, Cardoso and Kline, 2016) and Brazil (Morchio and Moser, 2024), race gaps in Brazil (Gerard et al., 2021), and immigrant-native wage gaps in Israel (Arellano-Bover and San, 2024). Card, Rothstein and Yi (2024) use similar LEHD data and models to explore drivers of racial wage gaps in the U.S. We build on these prior analyses by studying inequality both between groups and within-worker over time. Studying a time-varying worker characteristic also allows for new approaches to decomposing between-group earnings gaps, which the prior literature has primarily approached by setting within-firm pay gaps to zero in a low-surplus sector such as restaurants.

2 Sample construction and summary statistics

This section begins by describing the administrative criminal justice data, earnings records, and survey data used in this analysis. We then describe the measurement of a criminal record and sample construction. Lastly, we discuss summary statistics for workers with a criminal record and compare them to those of workers without any recorded criminal justice contact.

2.1 Data sources

Administrative criminal justice data. We use criminal justice data from two sources. The first is statewide court and corrections records from the Criminal Justice Administrative Records System (CJARS) (Finlay, Mueller-Smith and Papp, 2022). CJARS is a joint project of the University of Michigan and the U.S. Census Bureau to integrate and standardize criminal justice records from a variety of state and local sources. The second data source is statewide court and correction data we collected by sending individual requests to various jurisdictions across the U.S. All criminal justice data includes information on the date of offense, charges, and case disposition; the nature of the disposition itself (e.g., conviction vs. acquittal); and the dates of incarceration spells in state prisons. Both CJARS records and the data we collected were linked to an anonymous person code known as a Protected Identity Key (PIK) by Census staff on the basis of name, date of birth, and address information. PIKs can then be used to link individuals across Census data products.

Longitudinal Employer Household Dynamics (LEHD). The LEHD data consists of quarterly unemployment insurance earnings records shared with the Census by all fifty states and the District of Columbia (D.C.), covering 96% of private sector jobs (Abowd et al., 2009) and all state and local government workers. Federal employees, self-employed workers, members of the military, and some agricultural work are excluded. Projects must seek approval from individual states to access their LEHD data. Twenty-one states and D.C., accounting for about 45% of the total U.S. population, approved our request.[3] We also have access to a separate file that indicates whether an individual had earnings in *any* state, including those that did not approve the study, allowing us to construct an indicator for having any LEHD earnings nationally. LEHD staff have also created a companion file that provides age, sex, race, and education information for LEHD workers based on links to the Decennial Censuses, American Community Survey, and Social Security Administration records, with values imputed where necessary.

Firms in the LEHD data are identified by state employer identification numbers, which typically reflect the entity reporting UI taxes to state authorities and may comprise multiple establishments. LEHD data contain a separate quarterly earnings record for each worker-firm pair. We transform this data into a worker-level panel in each state by keeping the top-paying employer in each quarter as well as the sum of earnings from all employers and the number of employers. We inflate all earnings information to 2020 real dollars using the Consumer Price Index (CPI). The vintage of LEHD data we use covers employment from the 1990s through 2022, with exact start dates depending on the state. The records also contain information on several firm characteristics, such as North American Industry Classification System (NAICS) codes.

American Community Survey (ACS). We use ACS responses from 2001 to 2021 to construct several additional variables used in some of our analyses, such as occupations. Individuals are linked between ACS and LEHD data based on the date of ACS response and PIKs. When merging occupations, for example, we keep any ACS response within eight quarters of each quarterly LEHD observation.

2.2 Sample construction

Our sample is based on data from seven states where we both have access to LEHD records and where statewide court records cover a sufficiently long period to accurately measure

[3]These states are Arizona, Arkansas, California, Colorado, Delaware, Illinois, Indiana, Iowa, Kansas, Maine, Maryland, Montana, Nebraska, Nevada, New Mexico, North Dakota, Ohio, Oklahoma, Tennessee, Texas, and Wyoming, as well as the District of Columbia.

criminal histories. These states are Arizona, Arkansas, Iowa, Maryland, North Dakota, Oklahoma, and Texas. In each state, we include quarters where workers are aged 22 to 62 (inclusive) and exclude individuals born too early to measure criminal histories from age 18 on. We also excluded quarters after the criminal justice data ends in each state to ensure we cover charges and convictions over the full sample period. Because men make up the bulk of individuals in our criminal justice data, we drop women entirely. Finally, we drop all earnings records after 2019 to avoid the influence of the COVID-19 pandemic.

Table B.1 provides an overview of the cohort and period restrictions used in each state. Workers with any LEHD earnings in one of these seven states over the period covered and born after the required year will be included in the sample.[4] We then build a panel of quarterly earnings for each worker over the relevant sample window. Although each worker in the sample was originally observed in at least one of the seven sample states, we include earnings data from all 22 states where we have access as well as national employment indicators to capture any activity after migration.

We then attach criminal histories to each worker in the sample. We record the earliest felony conviction as the first quarter where the worker has a felony conviction disposed or is admitted to incarceration in the union of our criminal justice records and the CJARS datasets. We also consider a broader notion of criminal record status defined as having a prior felony charge (regardless of conviction) or a conviction for any lower-level crime excluding traffic and public order offenses. These two definitions of record status serve as our primary categorizations of interest. We refer to the former as the "any felony" sample and the latter as the "any record" sample. A worker is considered to have a criminal record of the relevant type in and after the earliest relevant quarter.

A key concern is whether our measures of criminal history accurately capture all charges and convictions for workers in our sample. We view the most likely type of error as cases where workers are incorrectly coded as having no record when they have one but failed to match to our data or acquired it in a jurisdiction where we do not have coverage. We mitigate this concern by using *all* criminal justice data in CJARS and our records to measure criminal histories, not just those from the seven states that form the basis of the analysis sample. This means prior convictions in states like North Carolina, Florida, and many other jurisdictions will still be captured in our data.[5]

Nevertheless, some mismeasurement is inevitable. To the extent that some workers we

[4] A small share of workers appear in multiple states. In these cases, we ensure they meet the sample restrictions in each of the states where they appear.

[5] The CJARS documentation provides a complete description of their data coverage.

classify as having no record in fact have one, we will likely understate the gaps between workers with and without a record. As we show below, however, the gaps we measure are enormous, suggesting that this issue is not too extreme. Out-of-state prior records are also relatively rare in our data. Approximately [*pending disclosure*]% of the workers in our sample have a felony conviction recorded outside the seven core sample states. About [*pending disclosure*]% of workers classified as having a felony record in our sample would not be if we used only criminal justice data from the state where they were first recorded as having any earnings.

Event-study restrictions: Our event study analyses in Section 7 use subsets of our core analysis sample with minor modifications. We extend the minimum age to 18 to observe any earnings since adulthood. Treatment and controls groups are then subsets of the main sample. For example, when examining felony convictions we consider individuals first convicted between ages 21 and 40 and control workers matched on demographic and other characteristics discussed further below.

2.3 Summary statistics

Table 1 presents summary statistics for all workers in our sample, those with a felony conviction vs. not ("felony" vs. "no felony"), and those without any type of record vs. not ("any record" vs. "no record"). The overall sample is 29 years old on average, 78% white, and is roughly split between having at least some college vs. none. About 8.2 million unique workers are present in the data, with employment at roughly a million distinct firms. About 15% of workers are recorded as having some criminal record and 6% as having a prior felony conviction. These means are lower than some prior work estimating the population prevalence of criminal records.[6] The difference likely stems from our focus on convictions instead of lower-level interactions such as arrests, which are less well measured in our data, our exclusion of traffic and public order offenses, as well as the requirement that all individuals in our sample have at least some interaction with the formal labor market that results in positive LEHD earnings.

Workers with a record tend to be less likely to be white and have lower levels of formal education, with differences more extreme in the felony vs. no felony sample. Workers with a record of any type also have significantly lower earnings and employment rates. On average, they earn $2,750 less per quarter. Log earnings, which condition on working, are 38 log points lower. Workers with a felony conviction face even worse labor market outcomes relative to

[6] Brame et al. (2012)'s analysis of the National Longitudinal Survey of Youth finds a self-reported arrest rate between the ages of 18 and 23 of 30% when non-response is treated as missing at random.

non-felons, earning $3,734 less per quarter and, conditional on employment, receiving wages that are 58 log points lower. A similar pattern appears in employment outcomes, with workers who have a felony record experiencing the lowest employment levels. These patterns underscore the substantial disparities associated with having a criminal record, with felony convictions linked to the most severe labor market penalties.

What drives these disparities in economic outcomes? One potential factor is the employment barriers imposed by a criminal record, particularly for those with felony convictions. Workers with records are more likely to be employed at firms where co-workers earn significantly lower wages, suggesting that these barriers push them into lower-paying jobs. Notably, differences in co-worker pay account for most of the earnings gap, indicating that differences in firm sorting may play a critical role in explaining wage disparities between workers with and without a record. Another contributing factor is differences in worker skills and preferences that were present before the worker acquired a criminal record. For instance, workers without records are twice as likely to hold a college degree, reflecting substantial gaps in skills and labor market productivity. While these descriptive statistics suggest that both pre-existing and post-conviction factors contribute to earnings disparities, their relative impact remains difficult to quantify. We next turn to an econometric model designed to help quantify the role of these different channels.

3 Econometric model

In this section, we propose a statistical model for quantifying the role of firms, pre-record characteristics, and changes in earnings potential post-record. Our analysis builds on two-way fixed effects models introduced by Abowd, Kramarz and Margolis (1999). Specifically, we model log quarterly wages, w_{it}, as:

$$w_{it} = \alpha_i + \psi_{j(i,t)} + \beta_0' X_{it} + G_{it}\left[\tilde{\alpha}_i + \chi_{j(i,t)} + \beta_1' X_{it}\right] + \epsilon_{it} \tag{1}$$

where $G_{it} \in \{0, 1\}$ is an indicator for worker i's record status in quarter t (e.g., whether they have a prior felony conviction), α_i and ψ_j are person and firm effects, $\tilde{\alpha}_i$ and χ_j are the interactions between person and firm effects and record status, and X_{it} are time-varying observables whose effects also depend on record status.[7] We estimate this specification separately for each of the different definitions of G_{it} of interest (e.g., any felony conviction,

[7] We use a third-degree polynomial in age and quarter-by-year fixed effects, all interacted with four levels of educational attainment. We follow Card et al. (2018) and Bonhomme et al. (2023) and normalize the age profile to be flat at 40. The overall level of the age profile is absorbed by the person effects.

any record).

This model amounts to the canonical AKM specification but fully interacted with record status. Holding the worker's firm fixed, log earnings can decline after acquiring a record because their person effect changes ($\tilde{\alpha}_i$), gaps in firm effects emerge (χ_j), or the influence of covariates shifts (β_1). Earnings may also be lower for workers with a record due to lower pre-record person effects (α_i), capturing differences in earnings potential that precede record acquisition altogether. Finally, workers with a record may tend to be employed at firms with different pay premia (ψ_j) both before and after they first acquire a record.

One interpretation of this model is as a purely statistical decomposition of earnings patterns for each group. A growing body of literature demonstrates that AKM models provides a good approximation to conditional means (see Card, Heining and Kline, 2013; Card et al., 2018; Kline, 2024), including recent work in the U.S. using Social Security Administration earnings records, tax records, and LEHD data from the Census Bureau (Song et al., 2019; Bonhomme et al., 2023; Card, Rothstein and Yi, 2023*a*,*b*). We show that the standard specification tests perform well in our setting too.

Interpreting firm effects as the *causal* effects of firms on earnings requires assumptions on the mobility processes that identify them. Assuming "exogenous mobility" (Card, Heining and Kline, 2013) rules out between-firm mobility correlated with unobserved determinants of earnings, which includes both shocks to workers' productivity as well as sorting based on match effects. Kline (2024) relates the assumptions required for a causal interpretation of firm effects to those used in difference-in-differences analyses. In the AKM context, a causal interpretation requires both parallel trends—so that outcomes in the absence of a move would have remained stable—and no selection on treatment effects—so that the wage changes of movers are a good guide to potential wage changes for the average worker. Whether these assumptions are satisfied remains contested (see, e.g., Borovičková and Shimer, 2024).

Other work has provided a model-based interpretation of Equation 1 (e.g., Lamadon, Mogstad and Setzler, 2022; Morchio and Moser, 2024). If employers set wages using standard Lerner markdown rules, for example, then log wages admit the representation in Equation 1 whenever markdowns are constant within firm and marginal product is log additive in worker components (constant and time varying), firm productivity, and any residual shocks to either (see, e.g., the model in Card et al., 2018). Identification of the parameters in such a wage equation would then depend on assumptions about the relevant unobservables. We do not take a stand on the model generating Equation 1 in this paper and primarily interpret it statistically, although we occasionally will refer to person effects and firm effects as related

to workers' and firms' productivities, respectively.

The model requires several normalizations. First, the mean of the firm effects ψ_j is not identified. How one normalizes these parameters is inconsequential for the analyses we consider, but for interpretability we follow recent work and set their mean to zero in the restaurant sector (NAICS 722), an industry commonly considered to be low-surplus (Card, Rothstein and Yi, 2023*a*,*b*, 2024).[8] Second, because the mean firm effect is not identified for either record status group, the mean gap in firm effects χ_j is also not identified. The normalization of these parameters matters since it determines the relative role of declines in person effects post conviction vs. the emergence of within-firm differences in pay premia. We follow the prior literature in considering normalizations based on an assumption of zero within-firm gaps in the restaurant industry (Card, Cardoso and Kline, 2016; Gerard et al., 2021; Arellano-Bover and San, 2024; Morchio and Moser, 2024),[9] as well as an alternative that calibrates differences in mean person effects post-conviction to explain sorting patterns across firms.

A third issue is that although both α_i and $\tilde{\alpha}_i$ are identified (up to normalization) for all workers observed with both $G_{it} = 0$ and $G_{it} = 1$, some workers acquire a record at an early age and are never observed with $G_{it} = 0$. For workers with $\min_t G_{it} = 1$, only $\alpha_i + \tilde{\alpha}_i$ is identified. This means that the overall mean of α_i given that $G_{it} = 1$ is also not identified. To solve this issue, we assume that $\Delta = E[\tilde{\alpha}_i | \min_t G_{it} = 0, G_{it} = 1] = E[\tilde{\alpha}_i | G_{it} = 1]$. Because Δ can be estimated directly along with $E[\alpha_i + \tilde{\alpha}_i | G_{it} = 1]$ using their sample analogues, we can back out $E[\alpha_i | G_{it} = 1]$ as $E[\alpha_i + \tilde{\alpha}_i | G_{it} = 1] - \Delta$.[10] While this procedure requires no selection bias due to conditioning on $\min_t G_{it} = 0$, we find that $E[\alpha_i + \tilde{\alpha}_i | G_{it} = 1]$ differs from $E[\alpha_i + \tilde{\alpha}_i | \min_t G_{it} = 0, G_{it} = 1]$ by only 0.008 log points when G_{it} is defined by having any prior felony and 0.016 log points when G_{it} is defined as having any record, suggesting any selection is minimal.

[8] Card, Rothstein and Yi (2023*b*) write: "This is a large sector, with 2.7% of national employment; is nontradeable, not geographically concentrated, and competitive; and does not require highly specialized skills. It is also typically one of the lowest-paying industries. This makes it a useful benchmark for other sectors." In our sample, the average firm effect is lowest in the accommodation / food service sector overall.

[9] Morchio and Moser (2024) derive their normalization from a theoretical model, which implies that firms exhibiting a zero within-firm gender pay gap must satisfy three conditions: (i) they rank near the bottom of the utility distribution for both genders; (ii) they treat men and women as almost perfect substitutes in production; and (iii) they offer little to no amenity value to the workers they employ. To meet these conditions, Morchio and Moser (2024) focuses on the smallest firms in the restaurant and fast-food sector.

[10] Computing these quantities requires estimating the model under a given normalization, but so long as the same one is used to both estimate the mean of $\tilde{\alpha}_i$ in the sample observed with both $G_{it} = 0$ and $G_{it} = 1$ and $E[\alpha_i + \tilde{\alpha}_i | G_{it} = 1]$ in the full population it is irrelevant because it affects both objects equally. We estimate the model in the full population and then compute the mean of $\tilde{\alpha}_i$ in the sub-sample observed both pre- and post-conviction using the same estimates, ensuring the normalization does not affect the results.

Finally, when estimating this model we follow the standard approaches for accounting for the fact that LEHD contains no information on hours (Sorkin, 2018; Card, Rothstein and Yi, 2023*a*,*b*). Specifically, we include only quarters with earnings of at least $3,800 (about full-time earnings at the federal minimum wage) from a single employer and drop the first and last quarter of each worker-firm spell. We also require a connected set for both record status groups. We use the union of the largest connected sets for both groups, dropping the small share of firms in the connected set for workers with a record but not without. Thus, although ψ_j is identified for every firm in the estimation sample, χ_j is only identified for every firm where at least one worker with a record has some employment.

4 Explaining earnings dispersion

We begin by using estimates of the parameters in Equation 1 to decompose sources of pay dispersion for workers with and without a criminal record. This decomposition is based on the idea that due to the additive structure of the model, earnings variances for each group ($G_{it} = 0$ vs. $G_{it} = 1$) can be written as:

$$Var_0(w_{it}) = Var_0(\alpha_i) + Var_0(\psi_{j(i,t)}) + 2Cov_0(\alpha_i, \psi_{j(i,t)}) + \eta_0 \tag{2}$$

$$\begin{aligned} Var_1(w_{it}) = {} & Var_1(\alpha_i + \tilde{\alpha}_i) + Var_1(\psi_{j(i,t)}) + Var_1(\chi_{j(i,t)}) + 2Cov_1(\alpha_i + \tilde{\alpha}_i, \psi_{j(i,t)}) \\ & + 2Cov_1(\alpha_i + \tilde{\alpha}_i, \chi_{j(i,t)}) + 2Cov_1(\psi_{j(i,t)}, \chi_{j(i,t)}) + \eta_1 \end{aligned} \tag{3}$$

where $Var_g(X)$ and $Cov_g(X, Y)$ are shorthand for $Var(X|G_{it} = g)$ and $Cov(X, Y|G_{it} = g)$, respectively, and the residual terms η_g capture dispersion due to covariates ($\beta_0' X_{it}$ and $\beta_1' X_{it}$) as well as unexplained earnings variation (ϵ_{it}).[11] These summaries of the joint distribution of firm effects, person effects, and within-firm gaps provide a high-level overview of the relative quantitative importance of elements of Equation 1 for pay dispersion within each group. Since this decomposition is focused on within-group earnings dispersion only, it is unaffected by any normalization choices. Because separately estimating α_i and $\tilde{\alpha}_i$ is possible only for workers observed both pre- and post-conviction, we focus in the record sample on the role of the "total" person effect $\alpha_i + \tilde{\alpha}_i$, which is identified for all workers ever observed with $G_{it} = 1$. The role of firms (ψ_j) and within-firm gaps (χ_j) in the record samples is of particular interest in our setting since these terms describe how important *where* a worker with a record is employed is for their relative earnings.

[11] Covariates in our setting capture the shape of experience profiles, which as noted earlier are normalized to be flat at age 40. Earnings levels at age 40 are reflected in person effects.

4.1 Person effects, firm effects, and within-firm gaps

Table 2 reports estimates of the variance and covariance of person and firm effects and within-firm gaps for workers both with and without a record and the resulting variance decompositions. Recent work has emphasized the importance of correcting for bias due to sampling error in these estimates (e.g., Kline, Saggio and Sølvsten, 2020; Bonhomme et al., 2023). Without correction, for example, variance estimates are biased upwards and covariance estimates can be biased downwards, a phenomenon sometimes referred to as "limited mobility" bias. In Table 2, we present both plug-in estimates and bias-corrected estimates constructed under a homoscedasticity assumption (Andrews et al., 2008).[12]

Panel A presents estimates treating G_{it} as an indicator for any felony record (Columns 1 and 2) and as an indicator for any criminal record (Columns 3 and 4). In the no-record sample, the bias corrected variance of firm effects (ψ_j) is approximately 5 log points, implying that moving one standard deviation in the firm effects distribution represents an approximately 22% change in earnings. These firm effects explain about 18% of the total variance of log wages, indicating a substantive role for firms in accounting for earnings dispersion. Person effects (α_i) are roughly twice as variable as firm effects, explaining 40% of the variance of log earnings for both workers without a felony and workers without any record. Finally, we find strong positive covariance between firm and person effects. Dividing the covariance by the square root of the product of the relevant bias-corrected variances yields a correlation coefficient of about 0.26. This pattern of higher earning workers sorting to higher earning firms explains a further 14% of earnings dispersion among workers without a record.

Our estimates for workers without a record are comparable to those found in other studies studying the overall population. For example, our uncorrected estimate of the covariance of worker and firm effects is 0.014, the same as in Table A1 of Card, Rothstein and Yi (2023*a*), who also use the LEHD. Moreover, our uncorrected estimate of 21% for the share of the variance explained by firm effects is similar to prior uncorrected estimates of the importance of firms that range between 15% to 25% (Card et al., 2018; Bonhomme et al., 2023; Kline, forthcoming).

Panel B of Table 2 presents estimates for workers with a felony record and those with any record. These findings show similar patterns to those in Panel A, with some key differences. First, firm effects ψ_j show a similar level of overall variance (5 log points), indicating that not all workers with a criminal record are concentrated in disproportionately low paying firms.

[12] Bonhomme et al. (2023) showed that the homoscedastic correction usually yields similar results to heteroscedasticity-robust alternatives (Kline, Saggio and Sølvsten, 2020) and has lower variability in Monte-Carlo simulations.

Because earnings dispersion is lower overall in the record samples, however, as a share of total variance firms explain slightly *more* of earnings dispersion for workers with vs. without a record.

Second, plug-in estimates of total person effects ($\alpha_i + \tilde{\alpha}_i$) indicate comparable variability to workers without a record, but our bias corrections have substantially more bite in this case. This occurs because, as we demonstrate further below, workers with a record tend to be employed much less frequently than workers without one, so their person effects are estimated less precisely and as a result are more over-dispersed. After bias correction, we find that person effects explain 26-33% of earnings dispersion in the record populations. A smaller role for person effects is consistent with lower average skill in this population. Differences in earnings due to accumulated schooling, for example, would impact a worker's person effects.

Third, although the marginal distributions of person and firm effects differ in some aspects between the two samples, they show a similar positive correlation. If anything, person and firm effects are more strongly correlated among workers with a record. This pattern of assortative matching explains about 13-14% of earnings dispersion. Thus, although there is significantly less dispersion overall in total person effects among workers with a record, relatively higher earning workers with a felony record or any record remain more likely to work at higher paying firms.

The final panel of Table 2 presents estimates of the dispersion of within-firm gaps in pay (χ_j). Before correcting for bias due to sampling error, these pay gaps account for approximately 12-18% of the variance in log earnings. Meaningful dispersion in pay gaps could indicate discrimination against workers with a criminal history in some firms. However, after correcting for over dispersion due to sampling error, the variance of within-firm pay gaps is effectively zero. In fact, the point estimate is slightly negative at -0.008 for workers with a felony record and -0.001 for workers with any record.[13] Because χ_j is only defined at firms where workers with a record are *ever* employed, this estimate pertains only to firms that are willing to hire at least one worker with a record. Nevertheless, the data are consistent with a world where any within-firm pay differences between workers with and without a record are common across all firms that employ workers with records.

These findings speak to a large literature that, following Card, Cardoso and Kline (2016), has used AKM models to study wage gaps between groups (Gerard et al., 2021; Gallen, Lesner and Vejlin, 2019; Dostie et al., 2023; Arellano-Bover and San, 2024). A common measure

[13]We also find a small positive covariance between gaps and firm effects and gaps and person effects after bias-correction, although given the low variance in gaps these estimates likely reflect noise as well.

of differential pay is the "rent-sharing" coefficient, defined as $1 + \frac{Cov(\chi_j, \psi_j)}{Var(\psi_j)}$. This coefficient can be interpreted as the projection of firm effects for workers with a record $(\psi_j + \chi_j)$ onto firm effects for workers without one (ψ_j). This statistic is typically motivated by a model where firm effects capture surplus sharing between firms and workers, but one group has lower bargaining power and therefore potentially captures a lower share (e.g., Card, Cardoso and Kline, 2016; Gerard et al., 2021). Before correcting for sampling error, we find rent-sharing coefficients of 0.62 and 0.87 for workers with a felony record and with any record, respectively. However, after correcting for sampling error, we find coefficients of 1.21 and 1.03, which are consistent with no compression in firm effects for workers with vs. without a record.

4.2 Validating AKM

Several papers have attempted to stress test how well the additive specification Equation 1 captures earnings patterns across workers and within the same worker over time. These tests have shown that in a variety of countries and contexts the model provides a good approximation to average log earnings across the firm and person effect distribution and for movers between firms. We replicate several key exercises supporting the additive structure of firm effects in our samples here. The results of our mover's analysis below, as well as the evidence from prior work (e.g., Card et al., 2018; Bonhomme et al., 2023), are consistent with the additive structure of the model and limited sorting on match effects.[14]

Figure A.1 presents event study plots of worker wages before and after transitions across firms with varying levels of co-worker pay, following the approach proposed by Card, Heining and Kline (2013). The plots show no evidence of pre-trends prior to the transition. Workers moving to firms with more highly paid coworkers experience wage gains, while workers moving to firms with lower paid coworkers experience wage losses. The post-move wage changes are symmetric: workers moving to firms with higher or lower co-worker pay experience wage gains and losses of similar absolute magnitude.

Figure A.2 reinforces this symmetry by plotting changes in wages for movers up the firm distribution against those for movers down the distribution. Consistent with the model's additive structure, the dots are clustered around a line with a slope of -1, indicating moves up vs. down offer equal and opposite changes in wages. The OLS estimate of the slope, which does not correct for attenuation bias due to sampling error in the right-hand side variable, is -.96.

[14] Borovičková and Shimer (2024), however, provide an example of a data generating process under which these tests do not reveal the presence of worker-firm match effects in earnings.

5 Explaining earnings gaps

In addition to explaining pay dispersion within groups, parameter estimates from Equation 1 can be used to assess sources of earnings disparities *between* groups. Specifically, the additive structure of our model implies that differences in average log wages can be expressed as:

$$\begin{aligned}\underbrace{E[w_{it}|G_{it}=0]-E[w_{it}|G_{it}=1]}_{\textit{raw gap}} &= \underbrace{E[\alpha_i|G_{it}=0]-E[\alpha_i|G_{it}=1]}_{\textit{person effects (pre-record)}} \\ &+ \underbrace{E\left[\psi_j|G_{it}=0\right]-E\left[\psi_j|G_{it}=1\right]}_{\textit{between firm sorting}} \\ &- \underbrace{\underbrace{E[\tilde{\alpha}_i|G_{it}=1]}_{\textit{change in person effects}} - \underbrace{E\left[\chi_j|G_{it}=1\right]}_{\textit{change in firm effects}}}_{\textit{change in ``earnings potential'' due to record}} \\ &+ \underbrace{\beta_0' E[X_{it}|G_{it}=0]-(\beta_0+\beta_1)' E[X_{it}|G_{it}=1]}_{\textit{covariates}}\end{aligned} \tag{4}$$

This expression allows us to decompose pay gaps into components attributable to several sources. The first component is differences in earnings potential that existed prior to record acquisition, as captured by mean pre-record person effects α_i. The next component captures differential sorting across firms with varying pay premia ψ_j. The next component reflects changes in person effects post record and mean within-firm pay gaps, the combination of which we dub changes in "earnings potential." Covariates explain the remainder; in what follows, we subtract off this term to focus on what explains earnings differences after accounting for experience profiles.

As noted in Section 3, a key challenge is that although the change in earnings potential is identified, the relative contribution of changes in person effects vs. mean within-firm gaps is not.[15] This normalization issue exclusively affects the third line of the decomposition. The contributions of pre-record person effects and between firm sorting are invariant to this normalization issue (as well as the overall normalization of firm effects ψ_j). We therefore begin our discussion with these components before discussing how to interpret changes in earnings potential.

[15] One can arbitrarily shift all changes in person effects by $-k$ while simultaneously adjusting pay gaps by $+k$, leaving fitted values unchanged.

5.1 Overall gaps

Table 3 presents the overall results of the decomposition in Equation 4. The log earnings gap between workers without a prior felony conviction vs. with one, net of covariates, is roughly 24 log points (columns 1 and 2).[16] Person effects, which capture gaps in earnings potential that existed *prior* to record acquisition, explain the majority of this gap (66%). As discussed in Section 3, this estimate is constructed by using the sub-sample of workers with a record observed both before and after their first conviction to estimate the average change in person effects pre vs. post record. We then subtract this estimate from the mean of total person effects in the full record population, or $E[\alpha_i + \tilde{\alpha}_i | G_{it}] = 1$, to produce an estimate of $E[\alpha_i | G_{it} = 1]$.[17] Columns 3 and 4 show that the gap between workers with any record vs. not stands at 17 log points, with person effects accounting for an even larger share of the total (77%) in this case.

Differences in firm premia, on the other hand, explain a comparatively small portion of gaps. They account for roughly 11% of the total earnings difference between workers with vs. without a prior felony conviction. Comparing workers with any record vs. not, we find a slightly smaller role for firm effects. As share of the total, firm effects explain 9% of the gap in this population. These estimates suggest some role for firms in explaining the labor market penalty of a criminal record, although smaller than what some previous analyses have found for the role of firms in explaining race or gender gaps. Gerard et al. (2021), for example, find that firm effects explain 18-20% of racial wage gaps in Brazil, while Card, Cardoso and Kline (2016) find they explain about 14-20% of gender gaps in Portugal. However, recently Card, Rothstein and Yi (2024) find that firm effects explain none of the racial gaps in the U.S.

Some of these sorting effects may come from differential access to particular industries with varying levels of average pay premia (Card, Rothstein and Yi, 2023*a*). Figure 1 illustrates the role of industry by plotting average firm effects (left y-axis) and the share of employment for each group (right y-axis) by 2-digit NAICS code. Industries are sorted on the x-axis based on their average firm effects from highest (mining, oil, and gas) to lowest (accommodation and food services).[18] Panel A shows that workers with a felony are disproportionately employed

[16] These are the gaps among full-time workers in "sandwich" quarters with earnings above the federal minimum wage, and so naturally smaller than the overall gaps in Table 1. We decompose gaps in the propensity to work full time later in Section 6.

[17] As noted earlier, because the choice of normalization for within-firm pay gaps impacts our estimate of the change in person effects pre vs. post conviction and the average total person effect equally, it does not affect our estimate of $E[\alpha_i | G_{it} = 1]$.

[18] Figure A.3 compares average firm effects by industry to those from Card, Rothstein and Yi (2023*a*), which use LEHD records from all states. Our estimates align closely with theirs.

in manufacturing, construction, administrative and support services, and accommodation and food services. As is clear from the figure, some of these industries turn out to be relatively high paying (e.g., construction), while others are not (food service). As a result, the *between* industry component of sorting is relatively unimportant, explaining just 11% of the total sorting effect. The bulk of sorting therefore occurs within 2-digit industry.

Panel B reports similar estimates for workers with any criminal record vs. no criminal record. Here, industry differences play a larger role in explaining sorting across firms and account for 51% of the gap in firm effects. The difference between Panels A and B mainly stems from manufacturing—workers with any record are less likely to be employed in this typically high-paying industry than workers with a felony record specifically. Of course, some pay differences across industries likely reflect compensating differentials (Sorkin, 2018; Humlum, Rasmussen and Rose, 2025). To the extent that construction jobs are high paying but difficult and physically taxing, for example, between-industry gaps in wages may understate between-industry gaps in utility.

The next row of Table 3 reports the portion of the log earnings gap between workers with vs. without a record not explained by either pre-record person effects or sorting across firms. This residual change in earnings potential reflects reductions in earnings at all firm and can be attributed in our model to either a drop in average person effects or the emergence of an average within-firm pay disparity. For both the felony and any record contrasts, the combination of these two factors is non-negligible, explaining roughly 23% of the total disparity in the former case and 15% in later. Relative to sorting effects, which capture how representation *across* firms contributes to gaps, changes in earnings potential, which capture common gaps *within* firms, are thus roughly twice as important.

The remaining rows of Table 3 decompose the change in earnings potential into its two components under several different normalizations. The first set of estimates, labeled "constant person effects," makes the assumption that $E[\tilde{\alpha}_i|G_{it} = 1] = 0$, implying that on average person effects are the same before and after acquiring a criminal record. This assumption therefore attributes all of the change in earnings potential to within-firm pay disparities. To the extent that person effects capture worker productivity, it implies that there is no direct effect of acquiring a criminal record on productivity. An arrest, charge, or conviction may, however, lead to skill depreciation due to time out of the labor force or coincide with other events that also reduce productivity (e.g., an injury or period of intense drug abuse). The next row of Table 3, labeled "zero mean gaps," allows for maximal productivity declines post-conviction by assuming that in fact within-firm gaps are mean zero. Given our estimate of zero dispersion in gaps in the record samples in Table 2, this assumption also implies that

two workers with the same person effects would be paid the same at *all* firms regardless of their record status. As a result, gaps are explained by either sorting or post-record person effects, with the vast majority (89%) accounted for by the latter.

The results under the zero mean gaps normalization happen to align closely with what we would find using alternatives proposed in the literature. For example, the next row of Table 3, labeled "zero gaps in restaurant sector", normalizes pay gaps to be mean zero in the restaurant sector (Card, Cardoso and Kline, 2016; Gerard et al., 2021; Arellano-Bover and San, 2024). This choice implies that, on average, workers with a record receive a small within-firm *bonus* of roughly 1 log point. While the restaurant sector is typically viewed as low-surplus and competitive, assuming zero gaps in this industry also rules out some plausible mechanisms by which a criminal record may reduce earnings. If workers with a record are disproportionately less likely to work in front-of-house jobs with access to the best tips, for example, within-firm gaps may emerge even among restaurants. Indeed, Agan and Starr (2018) finds large effects of disclosing a record on employer contact rates in a correspondence study of fast food and big-box chains, suggesting that record status matters to employers even in this sector.

The final rows of Table 3 consider a final alternative that we call "skill-based sorting." This normalization sets the gap in total person effects ($\alpha_i + \tilde{\alpha}_i$) such that it fully explains the gap in average firm premia in a model where mean firm premia ψ_j are linear in total person effects. Using the estimates in Table 2, we can construct an unbiased estimate of the slope of firm effects in total person effects in the non-record samples by dividing the bias-corrected covariance by the bias-corrected variance of person effects. Doing so results in a slope of 0.017. For workers with a felony record, that implies a gap in total person effects of $0.027/0.17 = 0.158$. This gap is very close to the pre-record gap in person effects, implying that change in earnings potential comes from within-firm gaps entirely. For the any record sample, the result is even more extreme, with differences in pre-record person effects over-explaining sorting and leading to implied increases in person effects post record.

Looking across these results, we reach several conclusions. First, it is clear that pre-record person effects are responsible for the majority of earnings gaps. This finding underscores the fact that individuals who interact with the justice system tend to have low levels of formal schooling and face a host of other challenges, many of which predate interaction with the justice system altogether. Second, sorting appears to contribute meaningfully to gaps, but is roughly half as important as sources of pay differentials within firms. Third, there is likely some role for within-firm pay gaps in explaining pay differences between workers with and without a record. In a world where differences in person effects explain all of

sorting (linearly), pre-record differences in person effects are sufficient to account for all of the observed sorting effects. Further declines in person effects would require *over* explaining the gap in firm premia due to sorting.

5.2 Conditional gaps

Comparing workers with a record to all workers without one may overstate gaps because of differences in education, age, and other factors. Table 4 presents decomposition exercises that account for various sets of observable characteristics. The methodology is identical to that in the previous sub-section, except that the non-record sample is re-weighted to match the observable characteristics of the record sample. We use characteristics that can be coded as a discrete set of indicators for membership in a particular covariate cell, so that the re-weighting procedure amounts to dropping cells where no workers have a record and weighting the non-record group by the relative share of workers with a record in each cell.[19] Panel A reproduces the unweighted, baseline gaps from Table 3 to facilitate comparisons.

Panel B adjusts for demographics by re-weighting based on the interaction of year of birth and four levels of educational attainment.[20] Consistent with workers with a record having less overall education, re-weighted log earnings gaps are smaller than in the baseline decomposition by about 2.3 log points for the felony contrast and 2.2 log points for the any record contrast. The reduction mainly comes from a smaller gap in pre-record person effects, which stands at 14 log points and accounts for 64% of the total for the felony vs. no felony comparison, and 11 log points (75%) for the any record comparison. Sorting effects are also slightly smaller in both cases, although they account for a similar share of the overall gap. The change in earnings potential is invariant to re-weighting the non-record sample because it involves expectations over the population with $G_{it} = 1$ only (i.e., it is already weighted to match the record group's observables). Because re-weighting reduces the overall gap, however, this final term explains a slightly larger share of the conditional disparity in Panel B.

Panel C re-weights based on cells defined by year of birth, education, two-digit industry, and city and state. Although one impact of having a record may be to influence where a worker lives and the industry in which they are employed, Panel B shows that substantial gaps in earnings remain within these cells—about 17 log points for the felony comparison and 11

[19] Because records are relatively rare, effectively zero cells have zero workers without a record and need to be dropped. In other words, covariate overlap is not an issue.

[20] These levels are the same used in estimates of Equation 1: less than high school, high school, some college, and bachelors or higher.

log points for the any record comparison. Pre-record person effect gaps shrink substantially after accounting for these observables, falling by roughly a third. Roughly two thirds of the gap in person effects between workers with vs. without a record thus reflects gaps within industry and place. Sorting effects, however, remain similar in magnitude to the overall sample, standing at 2.2 log points for the felony sample and 1.6 log points for the any record sample. As in Panel B, changes in earnings potential are not affected by re-weighting and so explain the same amount of log earnings differences as before.

Finally, Panel D presents the same exercise as Panel C but replacing industry with occupation-by-year codes from the ACS.[21] These results tell a similar story to those in Panel C for the comparison between workers with and without a felony. For the comparison between workers with any record vs. not, however, sorting effects become quite a bit smaller.

Taking these estimates together suggests several conclusions. First, industry and occupation are important for the raw earnings gap between workers with and without a record. Accounting for these factors reduces overall disparities by 25% or more. Second, the contribution of sorting remains important even within narrowly defined covariate cells, often explaining a similar fraction of gaps as in the unconditional decomposition. This suggests that if acquiring a criminal record shifts access to high paying jobs, the effect is roughly as strong among similar workers in similar types of jobs as between them. Finally, changes in earnings potential tend to increase in relative importance when conditioning on workers observables, but only because these observables absorb other potential driver of gaps while leaving the role of earnings potential unchanged.

6 Explaining employment disparities

The preceding analyses focused on explaining dispersion and between-group gaps in log earnings among workers employed full time. As demonstrated in Table 1, however, workers with a criminal record have substantially lower overall employment rates. For example, individuals with a felony record are 11 percentage points less likely to be employed and 18 percentage points less likely to hold a job over a full quarter that pays at least the federal minimum wage. While some of these gaps may reflect the labor market barriers faced by workers with a criminal record, they may also stem from differences in underlying skills

[21] We add these codes by attaching any ACS response within eight quarters of the focal quarter in the LEHD. We use the raw ACS occupation codes interacted with year because ACS codes change over time. Within-firm gaps change slightly relative to Panels A-C because the sample here contains only workers matched the ACS.

and productivity that pre-date contact with the criminal justice system. This section uses estimates from Equation 1 to explore the role of these two factors.

Figure 2 presents the key exercise. This graph is a binned scatter plot of average employment rates against person effects for workers with vs. without a felony record, with each bin representing a decile of the within-group person effect distribution. Individuals observed both before and after acquiring a record are treated as two different observations. The outcome is the average employment rate in "sandwich" quarters with sufficient earnings to qualify for inclusion in the sample used to estimate our primary specification Equation 1. The graph presents two versions of person effects for workers with a record. The first, dubbed "pre-record" effects, is an estimate of α_i constructed using the same strategy as in Section 5. The second, dubbed "post-record" effects, uses $\alpha_i + \tilde{\alpha}_i$ as workers with a record's person effect assuming that mean within-firm gaps χ_j are zero. The former can be thought of as the level of earnings potential workers with a record had prior to acquiring their record, while the latter can be thought of as earnings potential post-record (with differences stemming from either within-firm gaps or drops in person effects).

The results show that average employment rates are increasing in estimated person effects. The relationship appears fairly linear except in the right tail of the person effect distribution, where differences do not translate into differences in employment rates. Because person effects are noisily estimated, slope attenuation in the tails may simply reflect measurement error. Indeed, if the most extreme person effect observations reflect pure noise, one would expect a flatter slope in the tails. The lines on the figures present OLS slopes that are not corrected for attenuation bias due to sampling error. After correcting using the estimates from Table 2, the relationship is even stronger for both groups: employment rates increase with person effects at adjusted slopes of approximately 0.22 for the non-felony group and 0.61 for the felony group—roughly double the uncorrected slopes of 0.19 and 0.32, respectively.[22]

Key differences emerge between the felony record and non-felony population. First, at every level of estimated person effects, employment rates are higher for the non-felony group than the felony group. These gaps are quite substantial. At the median person effect for the record sample, for example, gaps in sandwich employment stand at 7-10 percentage points. While the size of this gap depends somewhat on the normalization chosen for the person effects, it is also clear under either polar case large conditional employment gaps remain.

[22]To correct for attenuation bias, we multiply the naive slope estimate from a regression of mean employment rates on mean person effects in each decile by the ratio of the unadjusted to bias-corrected estimates of the variances of person effects. This estimate is constructed separately for the record and non-record groups.

Indeed, mean post-record person effects for the felony group would need to be roughly 40 log points higher—implying 40% average within-firm gaps in pay—to equalize conditional mean employment rates.

Second, bias-corrected slopes tend to be larger for the felony record sample than the non-felony sample. This difference, which is unaffected by the normalization of mean within-firm pay gaps, appears to emerge because employment is more strongly increasing in person effects in the left tail of the distribution, where more mass is concentrated for the felony sample. This difference could reflect either a universal pattern about the returns to skill for employment or the fact that a felony conviction erects stronger employment barriers for relatively low- vs. high-skill workers. Indeed, Figure A.4 shows that person effects are strongly negatively correlated with incarceration rates, implying that workers with low person effects may be making more frequent and potentially disruptive transitions in and out of prison.[23]

To summarize the share of employment gaps explained by person effects, we use bias-corrected regressions of employment rates on person effects to conduct a simple Oaxaca-Blinder decomposition. This decomposition is based off a record status-group specific linear regression of average employment rates onto person effects such that if $G_i = g$ indicates an individual with record status g and $\bar{e}$ is their average employment rate, $\bar{e}_i = \gamma_g + \beta_g(\alpha_i + g\tilde{\alpha}_i) + \eta_i$. We bias-correct the OLS slope estimate for each group (and therefore the constants γ_g as well) using the signal-to-noise ratio for person effects from Table 2. The resulting decomposition is:

$$\begin{aligned} E[\bar{e}_i|G_i = 0] - E[\bar{e}_i|G_i = 1] = & \underbrace{\beta_0 \left(E[\alpha_i|G_i = 0] - E[\alpha_i|G_i = 1]\right)}_{\textit{pre-record person effects}} \\ & - \underbrace{\beta_0 E[\tilde{\alpha}_i|G_{it} = 1]}_{\textit{changes in person effects}} \\ & + \underbrace{E[\alpha_i + \tilde{\alpha}_i|G_{it} = 1](\beta_0 - \beta_1)}_{\textit{returns}} + \underbrace{\gamma_0 - \gamma_1}_{\textit{gap in constants}} \end{aligned} \tag{5}$$

The first term reflects the share of employment gaps attributable to differences in pre-record person effects. The second term captures the effect of changes in person effects post-record, which as noted above, we compute assuming zero within-firm gaps in for this exercise. The final terms reflect components unexplained by differences in person effects.

[23] Distinguishing between overall nonlinearities and record-specific returns to person effects for employment would require estimating the non-linear relationship between employment and person effects, which is difficult without access to additional instruments to correct for measurement error.

The results presented as text on Figure 2 show that pre-record person effects explain a meaningful portion of the employment gap: roughly 30%. Accounting for changes in earnings potential post record increases the share explained to 37%. However, the majority of the gap remains unexplained in both case, suggesting that individuals with a record continue to face significant barriers to employment.

Panel B of Figure 2 presents the same analysis as Panel A for workers with any criminal record vs. those with no record. The employment gap between these groups is roughly half the size of the gap in Panel A. Bias-corrected slopes are steeper for the any record sample than the non-record sample, although not as steep as the slope for the felony record sample in Panel A. Differences in person effects account for a larger share of the total employment gap in Panel B vs. Panel A. Still, substantial residual gaps—more than half of the total—remain.

7 The impacts of a criminal record

Our analysis has documented several key drivers of earnings and employment gaps between workers with and without a criminal record. However, it remains possible that the labor market penalty of a record is attributable to other factors that covary with record status instead of the record itself. Workers with a record may differ in their unobserved skills (e.g., aptitude for specific types of tasks), preferences, or race and other fixed characteristics, for example. These factors may also influence sorting and within-firm pay gaps. However, unlike many other worker traits, criminal record status is time-varying. This fact allows us to use the panel dimension of our data to ask how earnings and employment *change* after a worker acquires a record and what explains those changes.

To conduct this analysis, we use several sub-samples of the primary analysis dataset. The key "treatment" group consists of individuals convicted of a felony offense for the first time when aged 21 to 40. To ensure this event reflects the acquisition of the first serious criminal record, we exclude individuals with any prior charges. To measure changes in components of log earnings before and after the event, we also require each individual to have a quarter included in our estimation of Equation 1 both before and after the conviction. This ensures that each worker has an estimate of both pre-record person effects (α_i) and the change in person effects post-record ($\tilde{\alpha}_i$).

We then build several comparison groups for this sample. All comparison groups match each treated worker to a random control worker of the same race, born in the same year, and with the same level of education, but with varying levels of criminal justice contact. The

first comparison group requires control workers to have zero observed charges of any type. The second comparison group requires the control to be charged with but not convicted of a felony offense in the same calendar year as their treated match and never convicted of an offense in the future. The final comparison group requires controls to be charged with (and possibly convicted of) a felony at least five years after their treated match, capturing the typical event-study comparison of treated units to future-treated units. We then take the set of treated workers with a valid match across all three potential comparisons types. Due to how they are constructed, no worker can be present in multiple comparison groups, but we allow the same control to be matched to multiple treated units.

7.1 Effects on employment and earnings

Panel A of Figure 3 presents average earnings in quarters with no incarceration for each group over a six year horizon. For the felony conviction group, $t = 0$ is the quarter of conviction. For all other groups, $t = 0$ is the calendar year and quarter of their matched treated worker's conviction. Several key patterns emerge. First, earnings for the never-record group follow a smooth upwards trajectory over time as the sample ages. There are large gaps between the felony conviction group and the never record group well before conviction, underscoring that adjusting for demographics alone (age and education) does little to reduce observed labor market gaps, and suggesting that the never-record population may provide a poor guide to counterfactual outcomes for the felony convicted population. The future charges control group shows a similar pattern, although their earnings grow much more slowly, and the gap in average earnings at the start of the sample ($t = -12$) is substantially smaller. The difference between these two control groups reflects the fact that individuals with future criminal justice contact (but no record as of yet) are generally less attached to the labor market and earn less than demographically comparable individuals who never interact with the justice system.

By contrast, individuals convicted of a felony show a marked decline in earnings starting one year prior to conviction. Earnings reach their lowest point in the conviction quarter and then gradually recover. A similar though less sharp pattern appears for the charged-but-not-convicted group, suggesting that criminal prosecution alone can adversely impact labor market outcomes (Agan, Doleac and Harvey, 2023; Agan et al., 2024*b*). Consistent with negative selection into more severe forms of criminal justice contact, pre-conviction earnings are lowest among the felony-convicted group, followed by the charged group and the future charges group, while never-record population earns substantially more throughout the entire observation window.

Panel A of Figure 3 plots national employment rates in LEHD covered jobs for each group. The employment of the never-record control group increases over periods -12 to -4, likely capturing age profiles in employment, then stabilizes. The future charges groups also shows some initial increases before declining. The charged and convicted groups, however, show strong declines in the lead up to the focal event and depressed levels thereafter. The charged-but-not-convicted group, for example sees decreases of roughly 10 pp from two years prior to two years after conviction. In the felony conviction group, a large share of the declines for $t > 0$ are attributable to incarceration, which remains persistently elevated post conviction. Employment drops by roughly 10 pp overall but closer to 5 pp after dropping quarters with any incarceration.

To summarize the changes in employment and earnings for the felony convicted group relative to these comparison groups, we use a simple difference-in-differences estimator comparing periods at least one year prior to conviction ($t \leq -5$) to at least one year afterwards ($t \geq 5$). Excluding the two-year period around the conviction date itself reduces the influence of any transitory shocks surrounding the event. Table 5 presents the results. Comparing the felony conviction sample to the charged but not convicted and future charges groups, the estimates show reductions in employment of 5-7 pp, or 1-3 pp after dropping quarters with any incarceration. Overall earnings in quarters with no incarceration declines by \$250-530. The estimates indicate that the charging event itself can account for about 50% of the decline in earnings in non-incarcerated quarters relative to the future charges group. Consistent with the differential trends in the pre-period, effects on employment and earnings are substantially larger when using the never charged group as a control.

These estimates are broadly consistent with previous work analyzing the labor market consequences of criminal charges and convictions. Using criminal justice records from several states linked to Internal Revenue Service data, for example, Agan et al. (2024*b*) find declines in employment of roughly 6 pp around a felony charge that does not result in conviction and 10-17 pp after a conviction. As we find in our case, they argue some of these declines reflect subsequent criminal justice interactions (e.g., incarceration). Rose (2021) also documents large declines in earnings around a conviction for felony offenders in Washington State. Both papers also highlight the low levels of earnings and employment experienced by workers prior to the first criminal charge or conviction.

7.2 Effects on log earnings, firm effects, and earnings potential

Figure 4 further explores the impacts of a criminal record on labor market outcomes by examining patterns in log earnings in sandwich quarters, i.e., periods of full-time employment used to estimate our AKM specification earlier in the paper. Panel A presents patterns for log quarterly earnings. The felony convicted group shows trends similar to the charged-but-not-convicted and future charges populations several years prior the focal event. Earnings decline in the year or so before conviction and stagnate for a period afterwards. As a result, a sizable gap emerges comparing pre-conviction to post-conviction outcomes. Table 5 suggests an effect of roughly 0.03-.04 log points. The never-convicted group, by contrast, experiences faster growth than any of the other three populations long before the focal event. As a result, diff-in-diff estimates of log-earnings effects are twice as large. These patterns provide further support that the never-charged group provides a poor approximation to counterfactual outcomes for the felony convicted population.

The remaining panels in Figure 4 use estimates of Equation 1 to decompose changes in log wages into components attributable to firm effects and earnings potential, with the later defined as the sum of total person effects and average within-firm gaps. Specifically, Panel B shows average firm effects in the felony convicted group decline relative to both the charged-but-not-convicted and the future charges group pre vs. post conviction. The difference-in-differences estimate suggests impacts between 0.2 and 0.6 log points, depending on the control group. This finding implies acquiring a felony conviction leads to declines in average firm effects and that the sorting effects in Tables 3 and 4 do not merely reflect the influence of other worker characteristics correlated with record status that were present prior to acquiring the record. Although the impacts are significantly smaller than the sorting effects in Tables 3 and 4, as a share of the total log earnings decline relative to these two control groups the contribution is similar.

Panel B also shows that the never record population's average firm effect increases smoothly over the entire sample window. Relative to the estimates from Tables 3 and 4, sorting explains a slightly higher fraction (17%) of the log earnings gap between workers with a felony conviction and workers with no record in periods $t \geq 0$. Because these workers clearly experience differential trajectories in earnings and firm effects over the entire sample window, however, this naive estimate likely overstates the impact of felony record status on sorting. Indeed, the difference-in-difference estimates of sorting effects using the other two comparison groups implies sorting effects roughly 10-20% as large. Thus, only a small portion of the sorting effects documented in our log earnings decompositions likely reflects

the labor market barriers of a felony conviction record itself.

The two other factors that drive log earnings in our model are person effects and within-firm gaps. Due to the fact that the change in mean person effects post-record is not separately identified from the mean within-firm gap, we focus on how their sum evolves. This measure of earnings potential is important because it captures changes in changes in pay that occur *within* firm at *all* firms. Panel C presents the estimates. For the charged but not convicted and future charges group, we use estimates of Equation 1 that treat G_{it} as an indicator for having any prior felony charges so that person effects can also change around the first charging event (regardless of whether it results in conviction).

As Figure 4 and the estimates in Table 5 make clear, the estimated decline in earnings potential is 2 to 6 log points, depending on the comparison group. Declines are smallest relative to the charged group, who by construction experience a felony charge around the same time as the felony convicted sample. Interestingly, however, declines in earnings potential are clearly smaller after a felony charge vs. conviction. There are also gradual declines in earnings potential both before and after the conviction event for the convicted sample and the charged comparison group. Because these estimates condition on full, sandwich quarter employment, these patterns imply the composition of employed workers shifts over time towards lower earning workers. No such patterns appear for the never record and future charges comparison groups.

We omit a figure or estimates of the contribution of within-firm gaps because by definition they can only affect workers with a felony conviction or the charged but not convicted sample, making difference-in-differences style comparisons less useful. Because we estimate a variance of gaps close to zero, plotting mean gaps amounts to plotting the overall mean gap, which as discussed throughout the paper is not identified. Under the zero-mean gaps normalization used earlier, for example, all lines would fall on the horizontal axis at zero. Under alternative normalizations (e.g., the constant person effects assumption), gaps would jump up at zero but only for the felony convicted and charged groups.

Finally, it is possible that the additive specification in Equation 1 provides a poor fit to earnings patterns in the quarters surrounding a criminal charge or conviction. Figure A.5 assesses this question by plotting average residuals—i.e., log earnings minus estimated total person effects, firm effects, and gaps—for each group over time. Mean residuals are near zero throughout the observation window for the never-record and charged-but-not-convicted group. The felony convicted population shows negative average residuals in the year just before the conviction, consistent with transitory earnings shocks not captured in the model that

precede conviction. Interestingly, however, average residuals decrease steadily throughout the observation window for the future charges group, suggesting the simple static specification in Equation 1 provides a worse fit to earnings dynamics for workers who have a felony charge later in life, although the average residual remains small.

Taking stock of these results, we conclude that a naïve comparison of workers with a felony conviction to workers with no history of criminal justice interaction—even in a difference-in-differences framework—likely overstates the role the conviction itself plays in shaping labor market outcomes. Comparing workers with a felony conviction to two more credible control groups, we find declines in overall earnings, employment, and log earnings post conviction. Overall changes in *where* workers work explains a non-zero but small share of the overall decline. Thus, although the magnitudes differ, our conclusions are ultimately similar to those in Section 5 and 6: fixed characteristics of workers are responsible for the lion's share of the apparent labor market penalty of a criminal record. Acquiring a record does shift labor market outcomes, but primarily by shifting the overall level of employment and earnings potential at every firm, as opposed to driving workers into lower-paying firms.

8 Conclusions

Why do workers with a criminal record earn so little compared to similar workers without one? One plausible explanation is that having a record limits the availability of high-paying jobs. The labor market restrictions of a record are codified into law at the state or federal level, as well as in occupational licensing requirements. Other restrictions operate implicitly through employer discrimination motivated by concerns about legal liability, worker productivity, or other factors. Because many workers with a record face severe disadvantage from an early age, however, it is unclear how the jobs available to workers with a record differ from those they would have been able to access without one.

This paper studied this question using merged administrative earnings and employment and criminal history data. We find little evidence that job quality, as measured by firm pay premia estimated in two-way fixed effect models, explains earnings gaps between workers with vs. without a record. This finding holds in the cross-section comparing all workers with and without a record, as well as in a difference-in-differences framework that examines changes in firm premia after the first conviction relative to matched controls. Instead, the amount workers would earn at *any* firm declines sharply post-conviction. And conditional on earnings potential, workers with a record tend to be employed much less frequently.

These results imply that the labor market penalty of a criminal record primarily arises

through whether a worker finds a job at all and how much any firm would pay them, rather than where specifically they work. One interpretation is that criminal record restrictions are not tightly linked to job quality as measured by firm pay premia. Indeed, prior correspondence studies show that employers put significant weight on job applicants' criminal histories even in low value-added sectors such as fast food. While the industries that tend to employ workers with vs. without a record are meaningfully different, workers with records are also over-represented in some high premia industries such as construction. It is possible that after acquiring a record, workers predominately find work in jobs that pay roughly the same amount but have other unattractive amenities (e.g., outdoor physical labor).

The results also suggests that policies targeting overall employment and work in the types of jobs likely to hire workers with a record are more likely to decrease the record vs. non-record gap than policies seeking to reduce barriers in more prestigious and selective jobs these workers would be unlikely to access even without a record. To the extent that the post-conviction decline in earnings potential at every firm reflects decreases in productivity, the results point to a role for policy in mitigating skill depreciation around a criminal justice interaction. On the other hand, if these declines instead reflect pay discrimination or common patterns of within-firm job sorting, the results suggest that challenges in internal labor markets and pay policies, rather than barriers in overall access to the firm, are a key determinant of record penalties.

Tables and figures

Table 1: Summary statistics

	Full sample (1)	Felony (2)	No felony (3)	Any record (4)	No record (5)
Criminal justice interaction					
Any felony conviction	0.06	1.00	-	0.40	-
Any criminal conviction	0.15	1.00	0.09	1.00	-
Share of quarters with any incar.	0.01	0.10	0.00	0.04	0.00
Demographics					
Age	29.01	29.71	28.97	29.44	28.94
White	0.78	0.71	0.78	0.75	0.79
Black	0.13	0.23	0.12	0.19	0.11
Asian	0.04	0.01	0.05	0.01	0.05
Multiple races	0.02	0.02	0.02	0.02	0.02
Education					
Less than HS	0.19	0.25	0.19	0.24	0.19
Only HS	0.31	0.35	0.30	0.34	0.30
Some college	0.31	0.29	0.31	0.30	0.31
College or more	0.19	0.10	0.20	0.12	0.21
Labor market outcomes					
Total quarterly earnings	10,550	7,046	10,780	8,210	10,960
Log earnings	8.88	8.33	8.91	8.55	8.93
Share of quarters with any earnings	0.73	0.63	0.74	0.69	0.74
Share of quarters with earnings > $3800	0.61	0.44	0.62	0.53	0.63
Share of sandwich quarters	0.52	0.35	0.53	0.43	0.54
Average earnings of co-workers	10,450	8,088	10,600	8,869	10,730
Average quarters in sample	42.45	43.26	42.40	42.78	42.39
Number of workers ('000)	8226	646	7774	1360	7262
Number of firms ('000)	1097	349	1078	522	1046
Number of person-quarters ('000)	139000	8297	131000	20500	118000

Notes: This table presents summary statistics for the full sample (column 1), workers with a felony conviction vs. not (columns 2 and 3), and workers with any recorded criminal record vs. not (columns 4 and 5). All statistics are person-weighted. Demographic and education data are taken from the LEHD individual characteristics files. Labor market data refers to earnings and employment outcomes recorded in all states where this project has LEHD access. "Sandwich" quarters refers to quarters with earnings above $3,800 and with positive earnings from the same employer in the preceding and subsequent quarter. Average quarters in sample refers to the average total number of quarters over which each worker was observed between ages 22 and 62 in years where we have LEHD coverage (see table B.1 for details). All dollar values are inflated to 2020 equivalents using the CPI.

Table 2: Distributions of estimated person effects, firm effects, and within-firm gaps

	Felony record sample		Any record sample	
	Estimate (1)	Share of variance (in %) (2)	Estimate (3)	Share of variance (in %) (4)
A. No record				
$Var(w_{it})$	0.284	100.0	0.285	100.0
$Var(\psi_{j(i,t)})$	0.059	20.9	0.059	20.8
bias-corrected	0.052	18.4	0.052	18.1
$Var(\alpha_i)$	0.134	47.3	0.136	47.8
bias-corrected	0.115	40.4	0.116	40.8
$Cov(\psi_{j(i,t)}, \alpha_i)$	0.014	10.0	0.014	9.8
bias-corrected	0.020	13.8	0.020	13.8
B. Has record				
$Var(w_{it})$	0.233	100.0	0.249	100.0
$Var(\psi_{j(i,t)})$	0.058	24.9	0.060	24.0
bias-corrected	0.050	21.5	0.051	20.3
$Var(\alpha_i + \tilde{\alpha}_i)$	0.115	49.2	0.115	46.3
bias-corrected	0.061	26.0	0.081	32.6
$Cov(\psi_{j(i,t)}, \alpha_i + \tilde{\alpha}_i)$	0.016	13.9	0.017	13.3
bias-corrected	0.016	13.9	0.017	13.3
C. Within-firm gaps				
$Var(\chi_{j(i,t)})$	0.041	17.7	0.029	11.6
bias-corrected	-0.008	-3.5	-0.001	-0.5
$Cov(\psi_{j(i,t)}, \chi_{j(i,t)})$	-0.022	-18.8	-0.008	-6.1
bias-corrected	0.011	9.0	0.002	1.4
$Cov(\alpha_i + \tilde{\alpha}_i, \chi_{j(i,t)})$	-0.022	-18.8	-0.012	-9.5
bias-corrected	0.011	9.0	0.004	3.2

Notes: This table presents estimates of the variances and covariances of log earnings and firm and person effects from estimates of the model in Equation 1. The model is estimated separately treating G_{it} as an indicator for have a prior felony conviction (columns 1 and 2) and treating it as an indicator for having any criminal record (columns 3 and 4). All estimates are worker-quarter weighted. For variance estimates, the share of variance is defined as the variance of effects divided by the variance of log wages (e.g., $Var(\psi_{j(i,t)})/Var(w_{it})$). The share of the variance of the covariance terms is defined as twice the covariance divided by the variance of log wages (e.g., $2Cov(\psi_{j(i,t)}, \alpha_i)/Var(w_{it})$).

Table 3: Mean log earnings decompositions

	No felony vs. felony		No record vs. record	
	Estimate (1)	% of gap (2)	Estimate (3)	% of gap (4)
Log earnings	0.241	100%	0.168	100%
Person effects (pre-record)	0.159	66%	0.129	77%
Firm effects	0.027	11%	0.015	9%
Change in earnings potential	0.054	23%	0.025	15%
Constant person effects				
Change in person effects	0.000	0%	0.000	0%
Within-firm gaps	0.054	23%	0.025	15%
Zero mean gaps				
Change in person effects	0.054	23%	0.025	15%
Within-firm gaps	0.000	0%	0.000	0%
Zero gaps in restaurant sector				
Change in person effects	0.068	28%	0.028	17%
Within-firm gaps	-0.013	-6%	-0.003	-2%
Skill-based sorting				
Change in person effects	-0.001	0%	-0.042	-25%
Within-firm gaps	0.055	23%	0.067	40%

Notes: This table presents decompositions of the log earnings gap between workers without a felony conviction vs. with one (columns 1 and 2) and between workers with no record vs. any record (columns 3 and 4). Each decomposition is constructed using parameter estimates from Equation 1 applied to each type of record vs. no record comparison separately. Log earnings gaps are net of the estimated effects of covariates. Person effects refers to the first component in Equation 4 and captures differences in average person effects pre-record. This component is estimated by subtracting the average change in person effects pre- vs. post-record among workers where it is observed (i.e., $E[\tilde{\alpha}_i | \min_t G_{it} = 0, G_{it} = 1]$) from the average total person effect in the full record population (i.e., $E[\alpha_i + \tilde{\alpha}_i | G_{it} = 1]$). Firm effects refers to the sorting component in Equation 4, which captures differences in the employment share of workers across firm wage premia ψ_j. The change in earnings potential is the residual, which comprises either drops in person effects post record or increases in within-firm average pay gaps. The final rows of the table decompose this residual into its two components under different assumptions. "Constant person effects" assumes no change in person effects, while "zero mean gaps" assumes no average within-firm pay gaps. "Zero gaps in restaurant sector" assumes gaps are mean zero in NAICS 722 and implies a small average within-firm pay bonus for workers with a record. "Skill-based sorting" sets the average gap in total person effects ($\alpha_i + \tilde{\alpha}_i$) to explain the entire difference in average firm premia. This estimate uses a debiased estimate of the slope of firm effects on total person effects for the non-record sample computed using the bias-corrected variance and covariance estimates in Table 2.

Table 4: Conditional mean log earnings decompositions

	No felony vs. felony		No record vs. record	
	Estimate (1)	% of gap (2)	Estimate (3)	% of gap (4)
A. Baseline gaps				
Log earnings	0.241	100%	0.168	100%
Person effects (pre-record)	0.159	66%	0.129	77%
Firm effects	0.027	11%	0.015	9%
Change in earnings potential	0.054	23%	0.025	15%
B. Demographics re-weighted				
Log earnings	0.218	100%	0.146	100%
Person effects (pre-record)	0.138	64%	0.109	75%
Firm effects	0.025	12%	0.012	8%
Change in earnings potential	0.054	25%	0.025	17%
C. Industry, geography, demographics re-weighted				
Log earnings	0.169	100%	0.107	100%
Person effects (pre-record)	0.093	55%	0.067	62%
Firm effects	0.022	13%	0.016	15%
Change in earnings potential	0.054	32%	0.025	23%
D. Occupation, geography, demographics re-weighted				
Log earnings	0.158	100%	0.091	100%
Person effects (pre-record)	0.087	55%	0.067	74%
Firm effects	0.022	14%	0.006	7%
Change in earnings potential	0.052	33%	0.021	23%

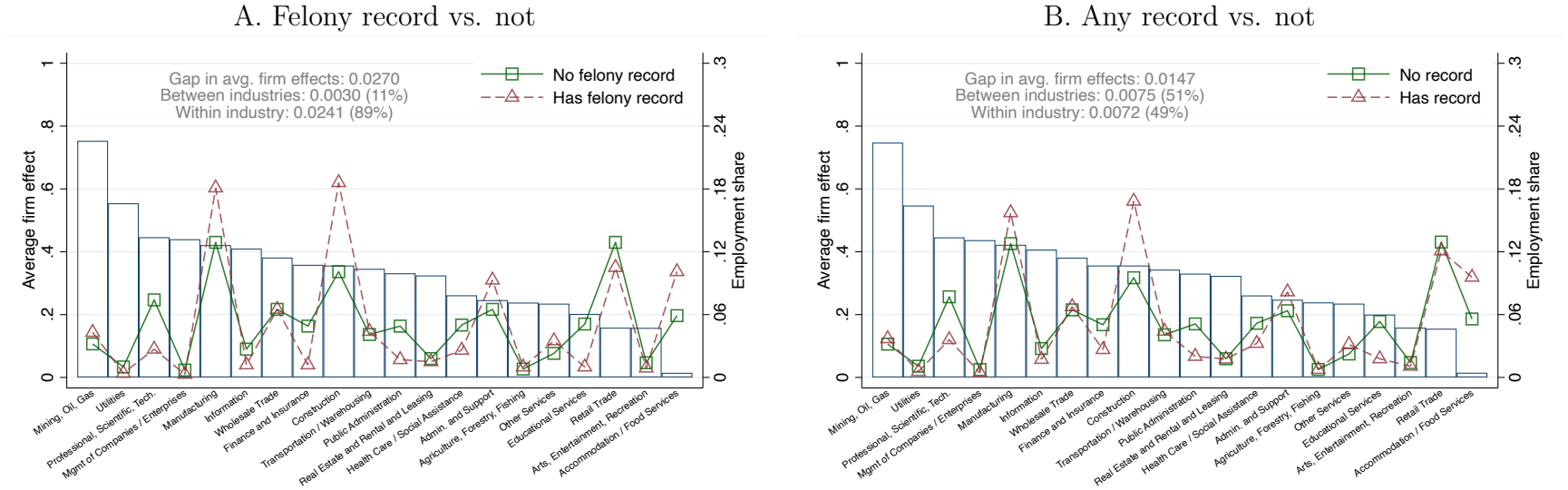
Notes: This table presents decompositions of the log earnings gap between workers without a felony conviction vs. with one (columns 1 and 2) and between workers with no record vs. any record (columns 3 and 4) after accounting for observable characteristics. The estimates are constructed in the same way as in Table 3, except the non-record group is re-weighted to mirror the record group's distribution of observables. Panel A reproduces the baseline gaps. Panel B re-weights based on the interaction of education and year of birth. Panel C adds the interaction of geography (Core-Based Statistical Area of employer) and two-digit industry, while Panel D uses occupation instead of industry. Occupation codes are added by matching the sample to proximate ACS responses (i.e., a response to the ACS within eight quarters of the relevant quarter in the LEHD). The sample in Panel D conditions on having a valid ACS match, and so is subsample of the full sample.

Table 5: Difference-in-differences estimates of the impact of a felony conviction

	Emp.	Emp. (not incar.)	Earn (not incar.)	Log earn	Firm FE	Earn. pot.
	(1)	(2)	(3)	(4)	(5)	(6)
Means pre-event						
Felony conviction	0.77	0.77	5,439	9.03	0.280	8.91
Never charged	0.83	0.83	7,479	9.15	0.287	9.09
Future charge	0.81	0.81	6,219	9.06	0.276	8.93
Charged but not convicted	0.80	0.80	6,007	9.06	0.285	8.94
Means post-event						
Felony conviction	0.67	0.72	5,876	9.11	0.281	8.84
Never charged	0.84	0.84	9,868	9.34	0.320	9.08
Future charge	0.78	0.78	7,182	9.18	0.283	8.93
Charged but not convicted	0.75	0.75	6,696	9.17	0.288	8.90
Differences post- vs. pre-event						
Felony conviction	-0.10	-0.05	437.50	0.08	0.001	-0.07
Never charged	0.00	0.00	2388.38	0.19	0.033	0.00
Future charge	-0.03	-0.03	962.88	0.12	0.007	0.00
Charged but not convicted	-0.05	-0.05	689.13	0.11	0.003	-0.04
DiD estimates						
vs. never charged	-0.10	-0.06	-1950.88	-0.11	-0.031	-0.06
vs. future charge	-0.07	-0.03	-525.38	-0.04	-0.006	-0.06
vs. charged but not convicted	-0.05	-0.01	-251.63	-0.03	-0.002	-0.02

Notes: This tables reports outcome means around a felony conviction for individuals convicted for the first time at age 21 to 40 and several matched comparison groups. The samples and outcomes are identical to those in Figures 3 and 4. All individuals used to construct the estimates are subsets of the sample used to estimate Equation 1. All comparison groups are matched on the level of education (four categories) and year of birth. The never-charged group has no recorded criminal justice interaction at any point. The future charge group was convicted of a felony at least five years after their matched convicted unit. The charged but not convicted group was charged with a felony but not convicted in the same year as their matched convicted unit and never convicted of a felony thereafter. Means pre-event refer to means over the period 12 to 5 quarters prior to conviction. Means post-event refer to means over the period 5 to 12 quarters post conviction. Earn. pot. refers to the sum of total person effects and average within-firm gaps $alpha_i + G_{it}[\tilde{\alpha}_i + \bar{X}_j]$. These two terms are not separately identified, although their sum is. Firm and person effects for the charged but not convicted and future charges groups come from estimates of Equation 1 treating G_{it} as an indicator for receiving any felony charges, allowing for changes in person effects and pay premia after an individual's first charge. See Section 7 for further discussion of sample construction and restrictions.

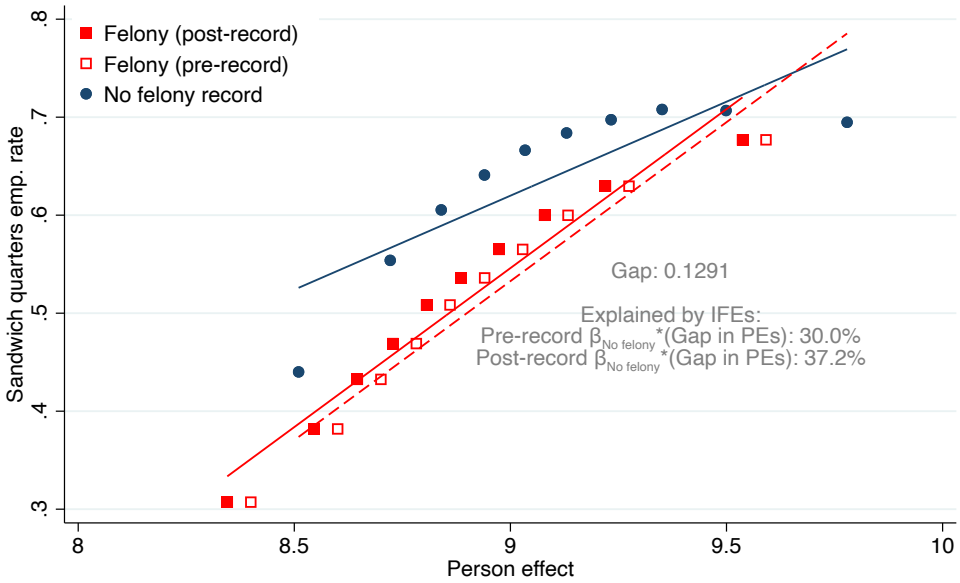
Figure 1: Industry wage premia and sorting patterns



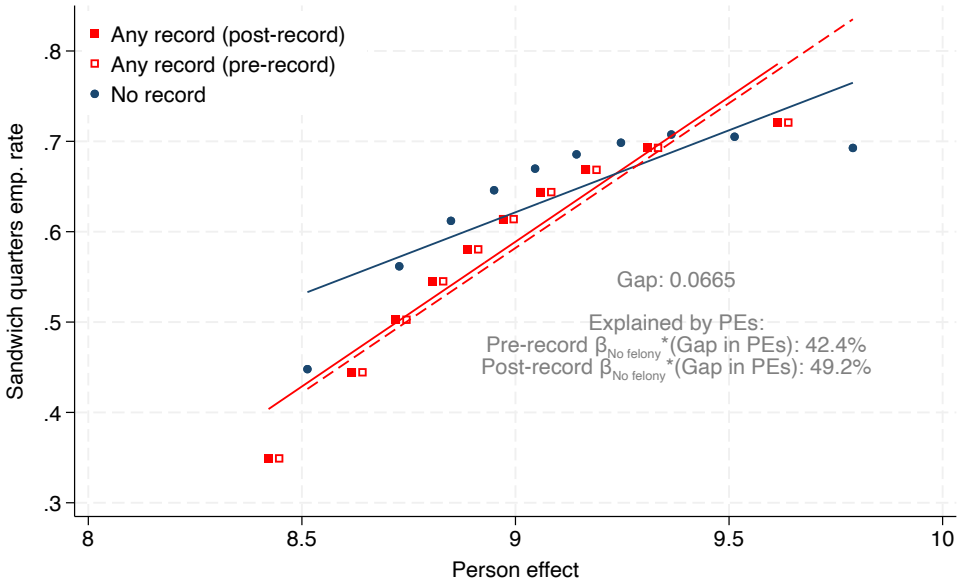
Notes: This figure presents industry average firm effects (ψ_j) from estimates of Equation 1. Industries are sorted from largest (left) to smallest (right), with their average firm effect (worker-quarter weighted using the non-record sample in each panel) on the left y-axis. The right y-axis reports the share of worker-quarter observations in each industry by group. Panel A reports results for workers with a felony record vs. workers without a felony record, while Panel B shows reports results for workers with any criminal record vs. those without a record.

Figure 2: Employment vs. person effects by criminal record status

A. Felony record

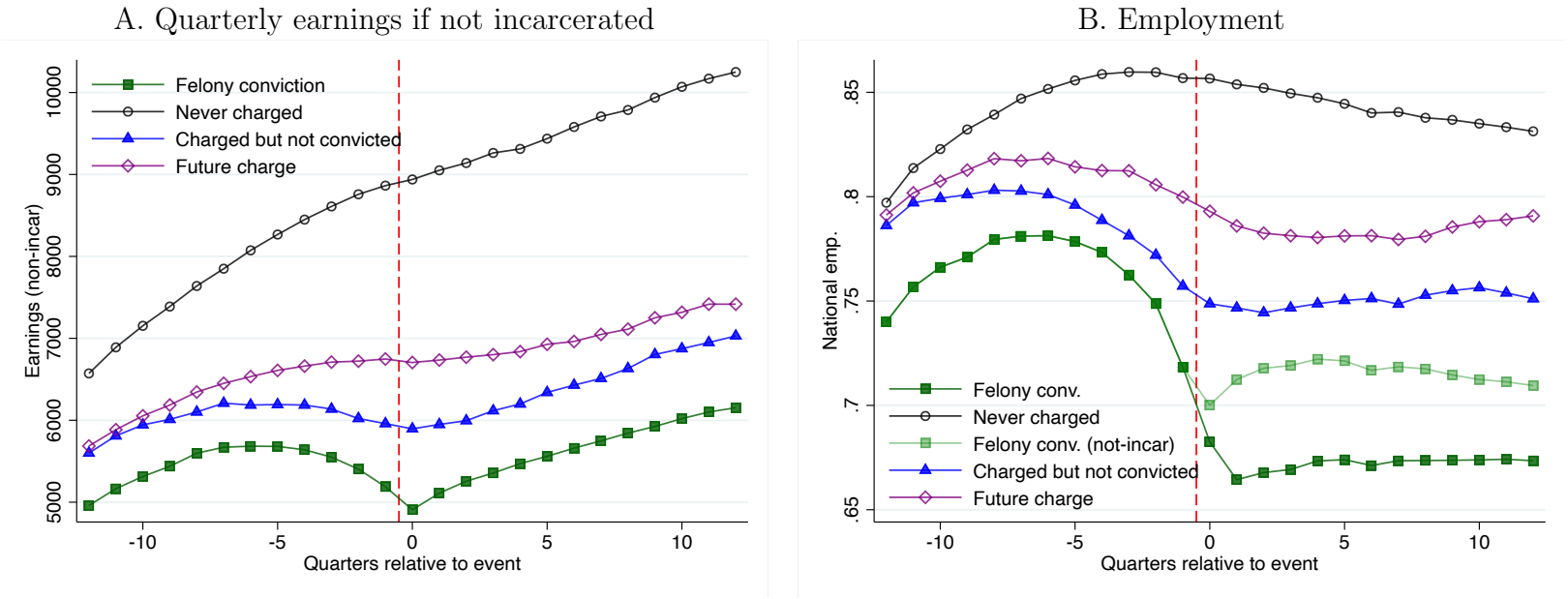


B. Any record



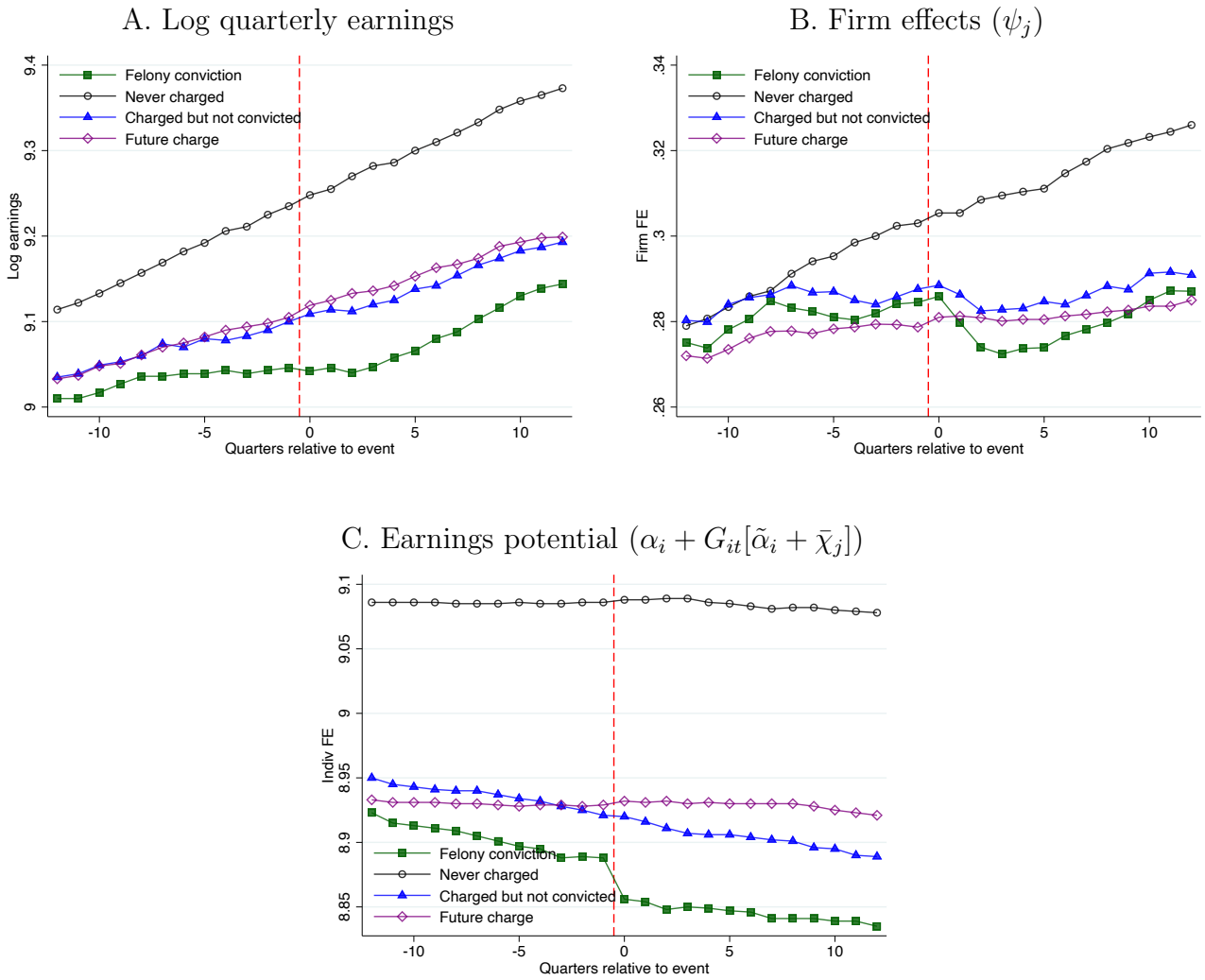
Notes: This figure plots average worker-level employment rates in "sandwich" quarters against estimated person effects by record status. A sandwich quarter is a quarter with earnings > \$3,800 and positive earnings from the same employer in the preceding and subsequent quarters. Panel A shows results for the felony record vs. non-felony record samples, while Panel B reports results for the any record comparison. Each dot reflects means after grouping workers into deciles based on their estimated person effects within each record group. For the record samples, the hollow dots capture pre-record person effects α_i computed using the strategy discussed in Section 3. The solid dots capture post-record person effects $\alpha_i + \tilde{\alpha}_i$ assuming that average within-firm pay gaps are zero. The lines are OLS slopes fitted to the reported means. The text reports the overall gap in employment rates and the share explained by the gap in person effects using the bias-corrected slope of employment on person effects for the non-record sample. The correction is computed by dividing the OLS slope by the signal-to-noise ratio of person effects from Table 2.

Figure 3: The dynamic effects of criminal records on earnings and employment



Notes: This figure presents earnings and employment dynamics around a felony conviction for individuals convicted for the first time at age 21 to 40 and several matched comparison groups. All individuals used to construct the figure are subsets of the sample used to estimate Equation 1. All comparison groups are matched on level of education (four categories), race, and year of birth. The blue "charged but not convicted" group was charged with a felony but not convicted in the same year as their matched convicted unit and never convicted of a felony thereafter. The purple "future charge" group was convicted of a felony at least five years after their matched convicted unit. The never-charged group has no recorded criminal justice interaction at any point. Event time on the x-axis refers to the date of felony conviction for the matched convicted unit. Panel A shows earnings in all states where we have LEHD access. Panel B shows the mean of an indicator for employment in all LEHD states. The light green line drops quarters where the felony convicted group has any recorded days incarcerated. By construction, none of the matched control groups have any incarceration over the sample window. See Section 7 for further discussion of sample construction and restrictions.

Figure 4: The dynamic effects of criminal records on log earnings components



Notes: This figure presents log earnings dynamics around a felony conviction for individuals convicted for the first time at age 21 to 40 and several matched comparison groups. The convicted sample and matched comparison groups are identical to those in Figure 3. The outcome in panel A is log quarterly earnings including only sandwich quarters with earnings above $3,800—i.e., quarters that qualify for inclusion in estimation of Equation 1. Panel B plots average firm effects from estimates of Equation 1, while Panel C plots average earnings potential, defined as the sum of total person effects and the average within-firm gap ($\bar{\chi}_j$). These two terms are not separately identified, although their sum is. Firm and person effects for the charged but not convicted and future charges groups come from estimates treating G_{it} as an indicator for receiving any felony charges, allowing for changes in person effects and pay premia after an individual's first charge. See Section 7 for further discussion of sample construction and restrictions.

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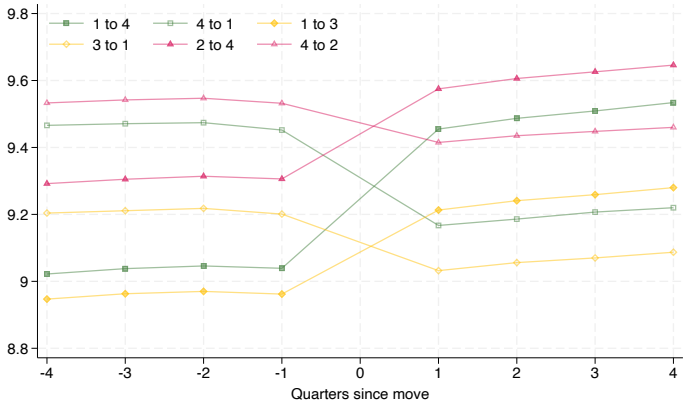
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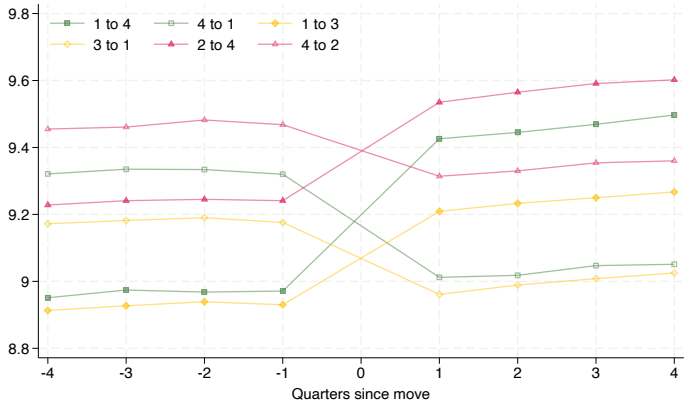
A Appendix figures

Figure A.1: Event studies around a worker's transition across firms

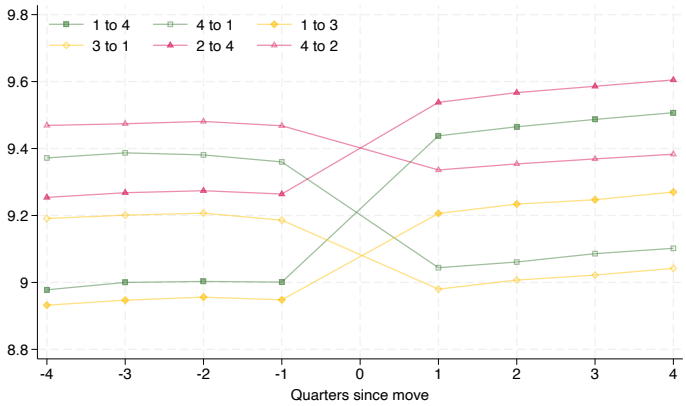
A. All workers



B. Workers with a felony record

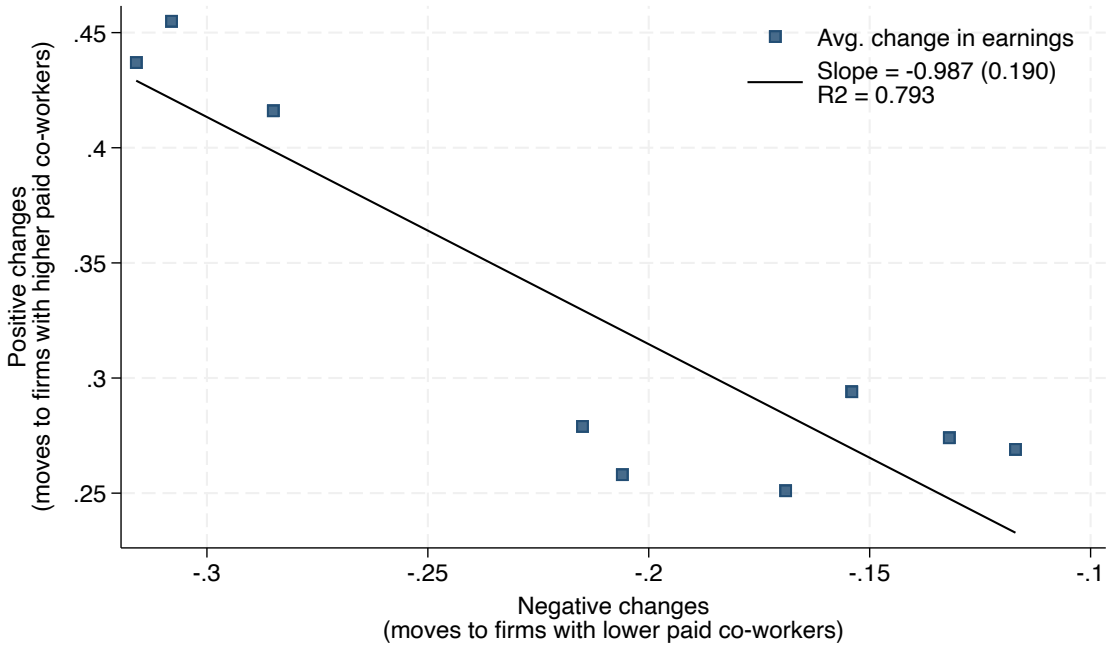


C. Workers with any criminal record



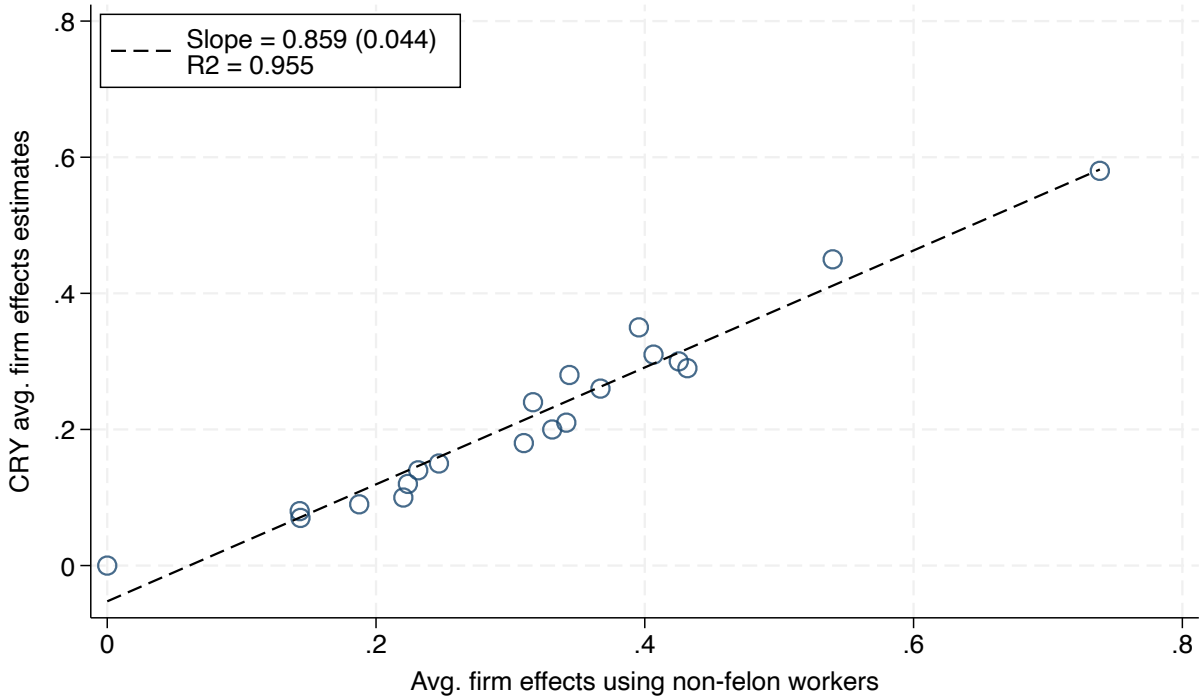
Notes: The figure plots the average log earnings of movers between firms classified by quartile of average co-worker log earnings. Quarter $t = -1$ refers to the last full quarter of employment (i.e., "sandwich" quarter) at the origin firm, while quarter $t = +1$ is the first sandwich quarter at the destination firm. We allow for up to four quarters of non-employment between these two employment spells. As is common in the literature (e.g., Card, Rothstein and Yi, 2023*b*), the movers sample is a subset of the full sample but restricted to workers employed at the same firm for at least four consecutive quarters before and after the move.

Figure A.2: Symmetry in moves between higher and lower paying firms



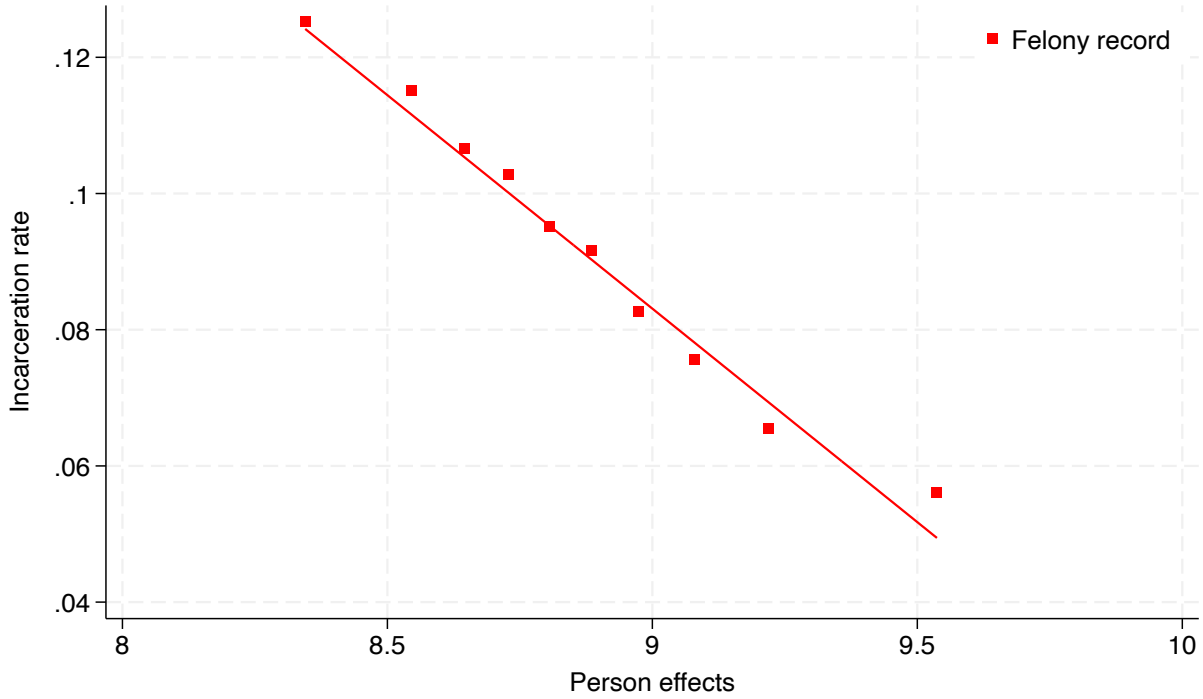
Notes: The figure plots the change in average log earnings at the time of the move (i.e., between quarters $t = -1$ and $t = 1$ in Figure A.1) of workers who move between firms, classified by the quartile of average co-worker log earnings at both the origin and destination firms. The figure includes estimates from all the groups (i.e., Panels A to C of Figure A.1). This figure visualizes the estimates in Figure A.1 differently to illustrate the symmetry in changes in earnings between switches upward to higher-paying firms and downwards to firms with lower average co-worker pay. Perfect symmetry implies all dots should fall on a line with a slope of -1, up to sampling error.

Figure A.3: Comparison of industry average firm effects to estimates from Card, Rothstein and Yi (2023*a*)



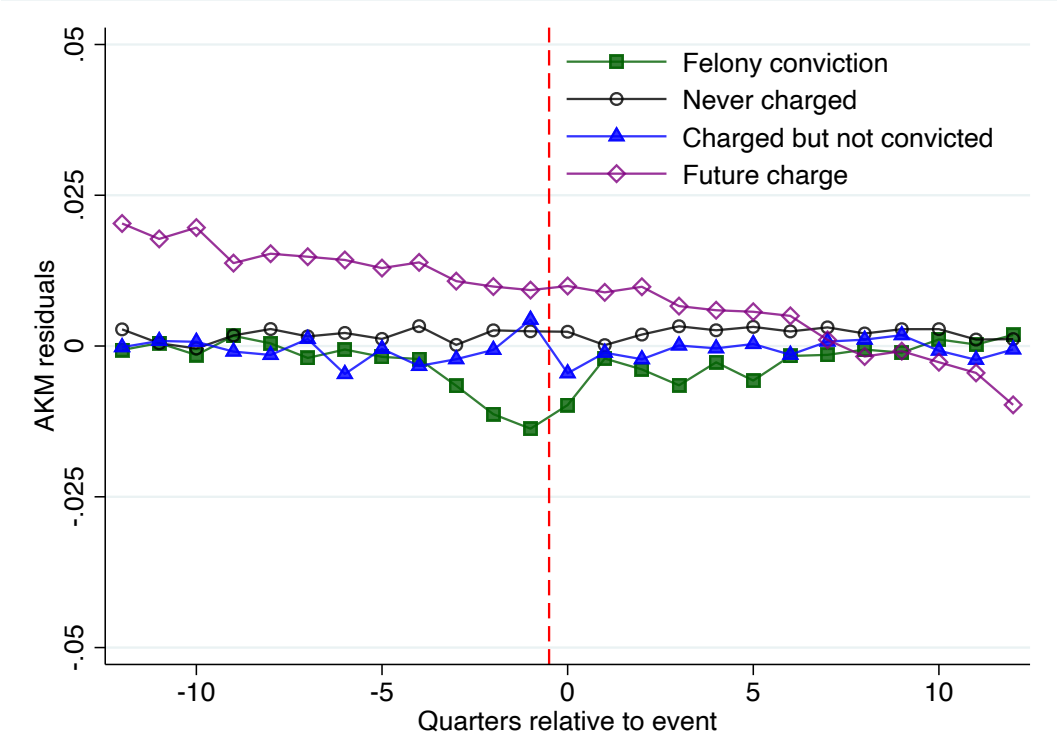
Notes: This figure compares our estimates of average firm effects for workers without a felony records by two-digit NAICS codes to estimates from Card, Rothstein and Yi (2023*a*), who estimate similar models using the full sample of LEHD states. All dollar values are inflated to 2020 equivalents using the CPI.

Figure A.4: The relationship between incarceration rates and person effects



Notes: This figure shows the relationship between estimated person effects and the share of quarters with any incarceration among workers with a felony record. The x-axis shows the average person effect in each decile of the person effects distribution. Person effects are computed assuming zero average within-firm pay gaps. The y-axis shows probability of having any incarceration in a given quarter.

Figure A.5: AKM residuals by event time



Notes: This figure presents average AKM residuals around a felony conviction for individuals convicted for the first time at age 21 to 40 and several matched comparison groups. The convicted sample and matched comparison groups are identical to those in Figure 3. The outcome is the residual from estimates of Equation 1 in the full sample. Estimates for the charged but not convicted and future charges group come from models treating G_{it} as an indicator for receiving any felony charges, allowing for changes in person effects and pay premia after an individual's first charge.

B Appendix tables

Table B.1: Sample description

State	Source of CJ data	Min. birth year	LEHD start year	LEHD end year
Arizona	Both	1978	2005	2019
Arkansas	Author collected	1974	2005	2018
Iowa	Author collected	1979	2005	2019
Maryland	CJARS	1974	2005	2017
North Dakota	CJARS	1977	2005	2019
Oklahoma	Both	1977	2005	2019
Texas	Both	1976	2005	2013

Notes: This table summarizes the cohorts and periods restrictions used for the seven LEHD approving states used to construct the analysis sample. Workers observed working over the period listed in one of these states and born after the listed minimum birth year are included in the sample. Workers observed in multiple states are required to satisfy the most conservative birth year restriction.